

Narrative Knowledge Weaver: Narrative-Centric Retrieval-Augmented Reasoning for Long-Form Text Understanding

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Abstract

Long-form narrative QA requires reasoning over evolving story worlds rather than isolated passages: answers may depend on earlier goals, changing character states, social relations, causal triggers, temporal position, and later consequences. Existing retrieval and graph-augmented generation methods improve evidence access, but their units—chunks, entities, relations, summaries, or tool actions—do not directly encode how evidence functions in a story. We introduce **Narrative Knowledge Weaver** (NKW), a source-grounded framework that aligns textual evidence, atomic facts, canonical graph structure, entity profiles, interactions, episodes, and storylines. At query time, NKW uses a snapshot-guided reasoner and skill-based reader to select source-grounded text, graph, and narrative evidence, then audit actor, scope, polarity, state, and temporal constraints; this closes the loop between the narrative fields materialized offline and the evidence operations selected online. Across STAGE, FairytaleQA, and QuALITY, NKW is strongest on screenplay-level story-world QA while remaining competitive on more passage-centered benchmarks. Ablations, question-type analyses, graph-asset statistics, and case studies show complementary benefits for character, scene, temporal, causal, and narrative-progression reasoning.

1 Introduction

Long-form narrative understanding supports screenplay analysis, production workflows, literary interpretation, and reading assistance. Unlike ordinary long documents, narratives are organized around scenes or chapters, recurring characters, changing states, and temporal-causal plot progression. Even with long-context LLMs, this remains challenging: the challenge is not only length, but recovering how evidence functions inside an evolving story world (Graesser and Franklin, 1990;

Kintsch, 1988; Zwaan and Radvansky, 1998; Liu et al., 2024).

Retrieval-augmented generation (RAG) partially addresses this challenge by grounding model responses in external evidence (Lewis et al., 2020; Izacard and Grave, 2021). This makes it useful for locating relevant passages and reducing unsupported generation in long documents. Graph-based RAG methods organize retrieved content through entity, relation, or community-level abstractions (Edge et al., 2024; Guo et al., 2025; Gutiérrez et al., 2024; Zhu et al., 2025), while tool- and retrieval-control RAG methods adapt the retrieval process during generation (Jiang et al., 2023; Trivedi et al., 2023; Asai et al., 2024; Du et al., 2026). However, these methods are largely developed for factual, open-domain, or general long-document settings, whereas narratives require different evidence organization.

We identify two mismatches between general-purpose RAG and narrative QA. First, narrative evidence is functional, not only locational: the same passage may serve as premise, turning point, consequence, state update, or distractor. For a why-question, retrieval may find the decision scene while missing the earlier motivating interaction. Chunks, entities, relations, and community summaries locate material, but do not encode its story role. Second, characters and relations are dynamic. Narrative questions ask what a character believes, wants, knows, or feels at a plot moment, or how attitudes change after a scene, requiring evolving states and relationships rather than a flat entity profile. Together, these mismatches make it difficult for general-purpose RAG systems to assemble evidence that is not only relevant in content, but also appropriate in narrative function and dynamic character context.

To this end, we present **Narrative Knowledge Weaver** (NKW), a narrative-centric RAG system for these two mismatches. To represent functional

evidence roles, NKW extracts source-grounded narrative assets and aggregates events, interactions, and occasions into higher-level episodes and storylines, so that evidence can be organized by its contribution to plot progression rather than by passage similarity alone. To capture dynamic characters and relations, NKW separates stable identity from changing state. It first links names, aliases, pronouns, and scene-specific references to the stable narrative entity. It then uses source-grounded atomic facts to build time-sensitive character profiles. Thus, characters are represented not as static graph nodes, but as evolving narrative states anchored in textual evidence. Furthermore, NKW provides a channel-separated tool interface over text, graph, and narrative views, together with skill manifests that guide evidence assembly and audit actor, scope, polarity, and temporal position at inference time. We evaluate NKW on three long-form narrative QA datasets, STAGE (Tian et al., 2026), FairytaleQA (Xu et al., 2022), and QuALITY (Pang et al., 2022), across seven LLM backbones against Hybrid RAG, GraphRAG, LightRAG, HippoRAG, and A-RAG. Results show the largest benefits when questions require reasoning over evolving states, relations, temporal order, causal motivation, and plot progression. Ablations, STAGE question-type breakdowns, graph-asset statistics, tool and skill inventories, and source-grounded case studies further show when the narrative structure contributes.

NKW couples offline narrative modeling with online evidence assembly: shared narrative fields organize indexed assets and guide skill-based reading, while source links preserve provenance. Matched controls and cross-model ablations evaluate this coupled design.

Our key contributions are:

- **A source-grounded narrative evidence architecture.** NKW couples canonical entity graphs, evolving profiles, functional narrative assets, and episode/storyline arcs while retaining a recoverable path to the supporting source.
- **An intent-conditioned evidence assembly loop.** A snapshot-guided reasoner selects the evidence view and one theory-aligned reading skill; the skill audits the source packet and may request a bounded supplement rather than triggering all retrieval channels by default.
- **Evaluation of the coupled design.** We evaluate NKW on STAGE, FairytaleQA, and QuALITY across seven LLM backbones, with matched

query-time controls, cross-model component ablations, chunk-size sensitivity, question-type analyses, and source-grounded case studies.

2 Related Work

Narrative Theory and Understanding. QuALITY, FairytaleQA, and STAGE evaluate narrative QA from passages to full screenplays (Pang et al., 2022; Xu et al., 2022; Tian et al., 2026). Narrative theory grounds this reasoning in situated change (Piper et al., 2021), story/discourse distinctions (Genette, 1983), and state transitions and disequilibrium (Bruner, 1991; Herman, 2009). Computational approaches model narrative structure and clause types (Ouyang and McKeown, 2014; Swanson et al., 2014), event chains and schemas (Chambers and Jurafsky, 2008, 2009), causal-temporal relations (Sun et al., 2024c; Mostafazadeh et al., 2016), event grounding (Jiayang et al., 2024), and turning points and arcs (Papalampidi et al., 2019; Boyd et al., 2020). Following the event-scene-plotline-plot organization reviewed by Piper et al. (2021), NKW uses three operational levels: source-grounded units, episodes, and storyline arcs. Scenes provide contextual views, while shared narrative fields guide offline modeling and online reading.

Structured Knowledge and Retrieval. Document-level extraction and coreference recover entities, relations, events, and normalized mentions (Yao et al., 2019; Wadden et al., 2019; Lin et al., 2020; Barhom et al., 2019; Cattani et al., 2020); EDC and Docs2KG extend graph construction with LLMs (Zhang and Soh, 2024; Sun et al., 2024b). GraphRAG, LightRAG, HippoRAG, and KG2RAG organize retrieval through graph structure (Edge et al., 2024; Guo et al., 2025; Gutiérrez et al., 2024; Zhu et al., 2025). RAPTOR builds recursive summary trees (Sarathi et al., 2024), HiRAG adds hierarchical knowledge (Huang et al., 2025), and DOS RAG preserves retrieved passage order (Laitenberger et al., 2025). NKW builds on structured evidence access by explicitly linking stable identities to changing states and local narrative units to episode/storyline progression.

Narrative Retrieval and Control. ComoRAG maintains a stateful memory workspace (Wang et al., 2025), NarCo models retrospective coherence links (Xu et al., 2024), and ChronoRAG restores adjacent chronological context (Kim et al.,

2026). A-RAG, RoG, ToG, and KG-Agent support adaptive retrieval or graph-grounded reasoning (Du et al., 2026; Luo et al., 2024; Sun et al., 2024a; Jiang et al., 2025); CRAG and CoV-RAG evaluate and revise evidence or answers (Yan et al., 2024; He et al., 2024). NKW couples functional narrative roles, evolving entity states, explicit episode/storyline progression, and narrative reading control. Its snapshot reasoner selects source expansions and one reading skill, which checks actor, time, state, and perspective and supplements unresolved evidence slots.

3 The Narrative Knowledge Weaver Framework

NKW models narrative understanding as reasoning over situated changes in agents, actions, goals, settings, and perspectives. It preserves the distinction between story time and discourse order while keeping every derived object linked to the source (Piper et al., 2021; Genette, 1983). Given source chunks $D = \{c_i\}_{i=1}^n$, NKW constructs $\mathcal{B} = (G, \mathcal{U}, \mathcal{F}, \mathcal{P}, \mathcal{H}, \mathcal{X})$, comprising a canonical entity–relation graph, source-grounded narrative units, atomic facts, entity-state profiles, episode/storyline structures, and provenance links. Construction is performed once; at query time, a snapshot reasoner selects evidence views and a skill reader assembles the answer (Figure 1).

3.1 Construction-Time Asset Building

3.1.1 Entity–Relation Backbone

NKW first builds a canonical graph over stable story referents. A chunk-level extractor emits entity records $v = (\text{name}, \text{type}, \text{desc}, S_v)$ and binary relation records $r = (u, v, \text{desc}, \text{keywords}, w, S_r)$, where source identifiers support later evidence recovery. The entity schema covers characters, groups, locations, time points, objects, institutions, roles, and concepts; narrative processes such as events and episodes are stored as separate typed units. N-ary statements are decomposed into binary edges, and missing endpoints can be represented by lightweight source-linked proxies. Same-name entities and same-pair relations are consolidated during upsert, while graph, vector, entity–chunk, and relation–chunk indexes remain synchronized.

3.1.2 Narrative Unit Extraction

In parallel with the graph backbone, NKW extracts three source-grounded unit roles. *Events* describe

actions, discoveries, decisions, arrivals, departures, or state changes; *interactions* describe directed communication, conflict, cooperation, or influence; and *occasions* describe stable situations, institutional constraints, social norms, background conditions, or scene-like contexts. Each unit records participating agents, action and target, stated goal or rationale, setting, story-time and discourse-order cues, narrator or focalization information, functional role, and source span when available. These fields are partial: an unstated goal, exact time, or perspective remains unknown rather than being filled in. The extractor also produces source-grounded atomic facts and initial entity attributes.

3.1.3 Entity Canonicalization

After graph merging, NKW resolves aliases and coreference so that local descriptions of the same character, object, place, institution, or plan share one canonical referent. The mapping updates participants, interaction endpoints, fact subjects and objects, and attribute owners without rewriting the natural-language evidence. Occasions are injected into the graph as contextual nodes linked to participants or settings, whereas events and interactions remain in the narrative store and vector index for retrieval and episode construction.

3.1.4 Entity Attribute and State Enrichment

NKW then selectively enriches salient entities. For entities with graph degree at least two in our reported implementation, it revisits linked source chunks and extracts entity-specific facts about states, actions, goals, causal roles, mental states, and persistent relations. Duplicate facts and overlapping attributes are merged into a compact attribute dictionary and semantic index. Attributes remain source-grounded and time-sensitive: before/after metadata and temporal cues preserve successive entity states, allowing the system to distinguish a character’s profile at the time asked from a later summary. The resulting profiles are used both in snapshot cards and in query-time entity/state retrieval.

3.2 Global Episode and Storyline Construction

The event–scene–plotline–plot organization reviewed by Piper et al. (2021), together with computational work on narrative structure (Ouyang and McKeown, 2014; Swanson et al., 2014), informs NKW’s three operational levels: narrative units,

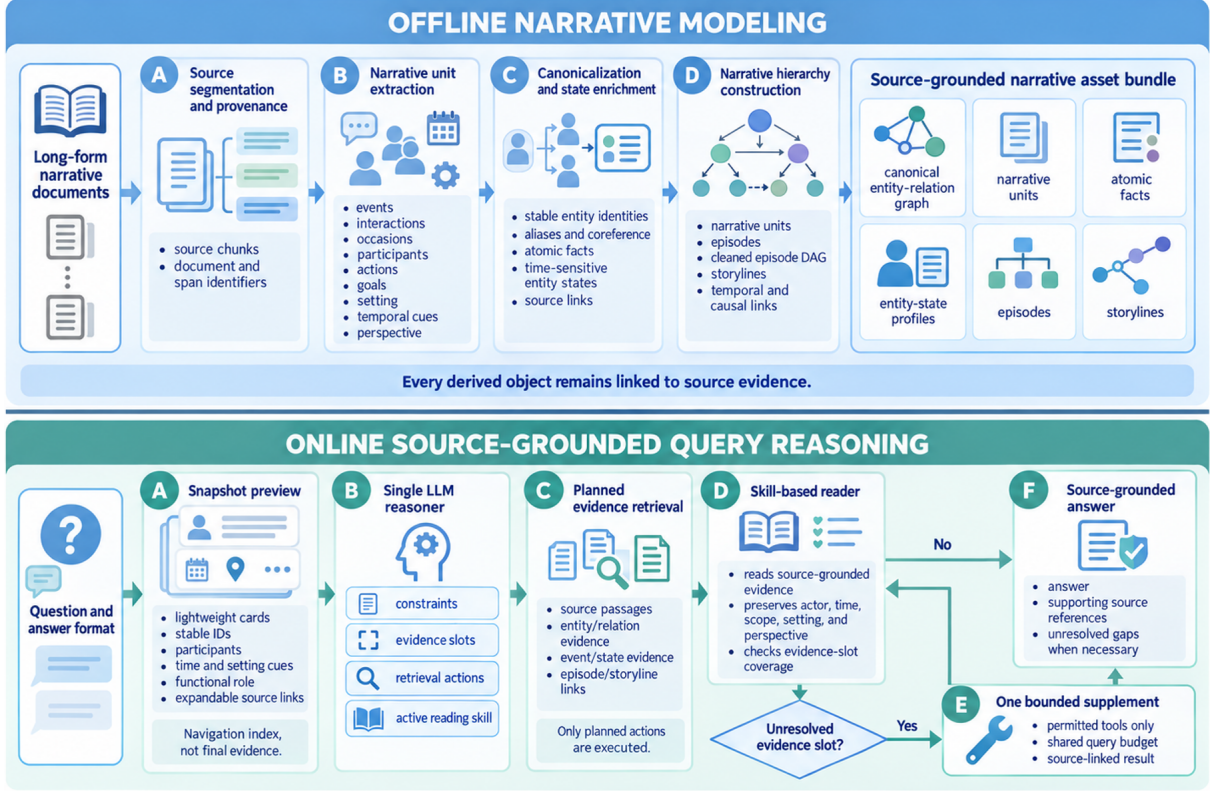


Figure 1: **Narrative-centric framework overview.** The framework separates construction-time narrative asset building from snapshot-guided query-time planning. A reasoner selects evidence actions and a skill-based reader may request only bounded, source-grounded supplements.

episodes, and storylines. Source-bounded regions provide scene-like context; episodes represent coherent progressions; and storylines capture longer plotline-oriented arcs.

The levels differ by narrative scope. A narrative unit records a local change or relation; an episode groups units in one developing situation or goal; and a storyline connects episodes into a longer trajectory with turning points and consequences. This operationalizes the reviewed narrative hierarchy in three queryable levels: units and episodes supply scene-like context, while storylines supply plotline-like structure. Fields move upward only with source support, so missing goals or outcomes remain unknown rather than being completed by a higher-level summary.

NKW groups events, interactions, and occasions into episodes using shared participants and goals, temporal and setting continuity, causal dependency, perspective compatibility, semantic similarity, and source support. Candidate construction is deterministic, with optional LLM adjudication and summarization after candidates are formed; answer labels are never used. NKW then predicts *causes*, *elaborates*, and *precedes* links between episodes using

source order, shared participants, graph neighbors, and episode content. SABER removes weak cycle edges and adjudicates redundant shortcuts only after the episode graph has an acyclic skeleton.

Storylines are extracted globally as trunk-branch episode chains. Their relations include blocking, resolving, prerequisite, conflict, and parallel arcs. Each storyline stores source-supported goals, obstacles, turning points, goal progression, observed outcomes, and conservative resolution status; child links keep every field expandable to its episodes and source passages. This representation follows narrative theory’s emphasis on change and turning points (Bruner, 1991; Herman, 2009; Papalampidi et al., 2019; Boyd et al., 2020) without requiring the system to materialize every theoretical level. Appendix A.1 details candidate construction, DAG cleaning, and chain extraction.

3.3 Narrative-Grounded Retrieval and Reasoning

3.3.1 Evidence Views

The asset bundle exposes complementary views over the same narrative: source passages for exact wording, atomic facts for compact propositions,

entity profiles for states and relations, event/interaction/occasion records for local structure, and episode/storyline links for long-range progression. Each view retains provenance. To prevent a salient entity from dominating the packet, query-time expansion attaches query-relevant attributes and the five highest-similarity atomic facts by default; the complete fact list remains available through an allowed entity operation.

3.3.2 Snapshot-Guided Query Planning

NKW first builds a lightweight *snapshot preview* rather than exposing the complete asset bundle to the reasoner. Four card families summarize canonical entities, atomic claims, typed narrative units, and episode/storyline threads. Cards contain stable identifiers, participants, time and setting cues, states, goals, functional roles, and expandable source/child links. They are a navigation index, not answer evidence: final claims must be supported by source passages or source-grounded records expanded from selected cards.

The preview also determines the first-pass granularity. For a question about why an agent changes state, entity/state cards establish the actor and prior state, unit/episode cards point to the trigger, and storyline links expose a consequence. The reasoner selects only the source spans and causal/state links needed to test those slots; literal questions stay local, whereas ordering and progression questions can expand higher-level links.

Given question q , answer format f , and preview Π_q , one LLM reasoner emits

$$P_q = (C_q, S_q, A_q, K_q), \quad (1)$$

where C_q specifies actor, scope, polarity, and temporal constraints; S_q contains evidence slots; A_q lists planned actions; and K_q selects one reading skill. The first retrieval pass executes only A_q , such as expanding source spans, inspecting an entity, or following an episode/storyline link. An unselected graph or narrative channel is not silently queried.

3.3.3 Skill-Based Evidence Assembly

The selected actions produce a compact source packet:

$$\begin{aligned} \mathcal{E}_q^{(0)} &= \text{Retrieve}(\mathcal{B}, A_q), \\ \mathcal{E}_q &= \text{Reader}_{K_q}(q, f, C_q, \mathcal{E}_q^{(0)}). \end{aligned} \quad (2)$$

The four skills are read-source for literal verification, track-character-state for identity and

changing states, trace-causes-and-outcomes for causal links and outcomes, and resolve-narrative-perspective for teller, focalizer, and attributed belief. Each skill defines a reading procedure and permitted tools. The reader checks the packet against C_q and answers when decisive slots are resolved. If a slot remains unresolved, it may request one bounded supplement round with at most two parallel evidence actions, tied to existing slots and a shared query budget.

The tool surface provides source wording (`find_source_evidence`), events and interactions (`find_event_evidence`), relations (`find_relation_evidence`), state transitions (`find_state_change`), temporal or causal chains (`trace_event_chain`), scene context (`locate_scene`), and canonical identity or attributes (`inspect_entity`). The final answer is generated from $\mathcal{E}_q$ with provenance attached to each admitted claim. Appendix A.2–A.4 gives the full card schema, evidence-binding rules, tool permissions, and skill procedures.

4 Experimental Setup

Datasets. **STAGE** (Tian et al., 2026) contains 151 movie screenplays (109 English and 42 Chinese) and 5,010 questions across six narrative reasoning types (Appendix B.5). **QUALITY** (Pang et al., 2022) contains 162 English passages and 2,523 multiple-choice questions; **FairytaleQA** (Xu et al., 2022) contains 278 children’s stories and 10,580 expert-authored questions. Mean document lengths in our processed benchmarks are 27,546, 4,115, and 2,165, respectively, measured in whitespace-delimited words for English and characters for Chinese.

Baselines. We compare with **Hybrid RAG**, **GraphRAG**, **LightRAG** (Guo et al., 2025), **HippoRAG** (Gutiérrez et al., 2024), **A-RAG** (Du et al., 2026), and **DOS RAG** (Laitenberger et al., 2025), covering non-graph retrieval, graph-based retrieval, agent-controlled retrieval, and ordered passage reading.

Implementation. Main comparisons use **600-token chunks**; Appendix B.9 examines 400, 600, 800, and 1200 tokens on a stratified 40-screenplay subset. Backbones include Qwen3 in non-thinking mode, Llama-3.1 Instruct, and GPT-5.5 as a closed-source reference. Qwen3-30B denotes **Qwen3-30B-A3B-Instruct-2507**. Ablations remove one

Table 1: Main QA results on STAGE across backbones. Within each backbone row, bold marks the best method separately for the Overall columns and for the Pass@5 columns.

Backbone	Hybrid RAG		GraphRAG		LightRAG		HippoRAG		A-RAG		DOS RAG		NKW (Ours)	
	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5
Qwen3-8B	0.422	0.464	0.285	0.312	0.368	0.433	0.264	0.289	0.232	0.274	0.332	0.365	0.385	0.501
Qwen3-30B	0.514	0.556	0.303	0.413	0.387	0.539	0.319	0.344	0.298	0.376	0.415	0.452	0.589	0.664
Qwen3-235B	0.606	0.646	0.321	0.479	0.441	0.599	0.401	0.430	0.362	0.486	0.505	0.549	0.701	0.815
Llama-3.1-8B	0.385	0.412	0.292	0.325	0.342	0.411	0.272	0.293	0.215	0.253	0.311	0.342	0.372	0.485
Llama-3.1-70B	0.512	0.555	0.312	0.469	0.391	0.522	0.410	0.442	0.335	0.421	0.421	0.462	0.563	0.643
Llama-3.1-405B	0.611	0.638	0.352	0.498	0.453	0.589	0.487	0.511	0.395	0.502	0.519	0.565	0.653	0.743
GPT-5.5	0.622	0.652	0.402	0.536	0.478	0.612	0.512	0.543	0.422	0.532	0.531	0.582	0.698	0.811

component at a time, matching the corresponding full system’s backbone, decoding configuration, questions, metrics, and query-time reasoning budget. Question-type ablations use **Qwen3-235B**; core component ablations additionally cover Qwen3-8B and Qwen3-30B (Appendix B.7).

Evaluation. DeepSeek-V4 evaluates STAGE and semantic-answer FairytaleQA responses against the question and ground-truth answer; QuALITY uses exact option matching. **Overall** denotes each benchmark’s primary correctness metric; **Pass@5** measures whether any of five sampled answers is correct.

5 Experimental Results and Analysis

5.1 Main Results

STAGE. NKW achieved the highest Pass@5 across all seven backbones and the highest Overall accuracy on five of seven (Table 1). With Qwen3-235B, Overall reached 0.701, compared with 0.606 for the strongest baseline, Hybrid RAG. The two 8B backbones were the exceptions: Hybrid RAG led on Overall, while NKW retained its Pass@5 advantage.

FairytaleQA and QuALITY. NKW led on Pass@5 for five of seven backbones on each benchmark (Tables 2 and 3). Overall performance varied more: DOS RAG led on three FairytaleQA backbones, while NKW led on four QuALITY backbones. LightRAG achieved the best Pass@5 for both 8B backbones on both benchmarks.

Across the three benchmarks, NKW’s lead was most consistent on screenplay-level QA and Pass@5. The smaller-model results show that these gains do not always extend to single-answer accuracy. Appendix B.5 reports the STAGE question-type breakdown, and Appendix B.8 presents representative successes and failures.

5.2 Controlled Comparisons and Ablations

Prompt- and budget-controlled comparisons. NKW retained an advantage when the retrievers shared the same online controller (Table 4). On the full STAGE set with Qwen3-235B, all four variants used the same reasoner, skill-reader and answer prompts, decoding configuration, maximum final-evidence budget, and one-round supplement protocol. The baseline variants changed the evidence provider while retaining this controller.

Full NKW achieved 0.701 Overall versus 0.628 for HybridRAG + Agent + Skills, with fewer mean evidence tokens (7,210 versus 8,140) and supplement actions (1.15 versus 1.45). Removing narrative assets reduced Overall to 0.648 and Pass@5 from 0.815 to 0.742. These controls support a contribution from narrative assets under a common controller and budget cap.

Core component ablations. Removing any of the five components reduced both metrics on all three benchmarks (Table 5). Among the offline components, removing narrative hierarchy lowered Overall by 5.3 percentage points on STAGE and 7.6 on QuALITY, compared with 1.5 on FairytaleQA. The online components also contributed: removing the snapshot-guided reasoner lowered STAGE Overall by 6.0 points, and removing skill-based evidence assembly produced the largest drop, at 7.6 points.

The STAGE question-type ablations further distinguish these effects (Appendix B.6): attribute removal most affected character and role-relation questions, while hierarchy removal most affected temporal reasoning.

Ablations across model sizes. All five components also improved both metrics with Qwen3-8B and Qwen3-30B (Appendix B.7). Removing skill-based evidence assembly caused the largest Overall loss at every size: 7.0, 6.9, and 7.6 percentage

Table 2: QA results on FairytaleQA. Bold marks the best method separately for Overall and Pass@5 within each backbone row.

Backbone	Hybrid RAG		GraphRAG		LightRAG		HippoRAG		A-RAG		DOS RAG		NKW (Ours)	
	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5
Qwen3-8B	0.732	0.746	0.312	0.365	0.679	0.823	0.771	0.813	0.387	0.473	0.715	0.745	0.652	0.801
Qwen3-30B	0.811	0.824	0.395	0.463	0.856	0.924	0.801	0.812	0.470	0.530	0.838	0.862	0.852	0.945
Qwen3-235B	0.841	0.850	0.449	0.525	0.875	0.933	0.826	0.839	0.587	0.703	0.892	0.911	0.875	0.955
Llama-3.1-8B	0.722	0.731	0.334	0.390	0.716	0.772	0.708	0.720	0.425	0.521	0.702	0.731	0.684	0.751
Llama-3.1-70B	0.813	0.825	0.464	0.541	0.801	0.834	0.796	0.804	0.553	0.668	0.811	0.835	0.819	0.861
Llama-3.1-405B	0.849	0.843	0.552	0.615	0.862	0.894	0.854	0.870	0.642	0.746	0.874	0.902	0.865	0.921
GPT-5.5	0.839	0.855	0.585	0.642	0.893	0.921	0.871	0.886	0.672	0.785	0.901	0.924	0.884	0.941

Table 3: QA results on QuALITY. Bold marks the best method separately for Overall and Pass@5 within each backbone row.

Backbone	Hybrid RAG		GraphRAG		LightRAG		HippoRAG		A-RAG		DOS RAG		NKW (Ours)	
	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5
Qwen3-8B	0.552	0.559	0.513	0.613	0.704	0.802	0.535	0.621	0.503	0.616	0.713	0.724	0.684	0.781
Qwen3-30B	0.622	0.628	0.593	0.692	0.753	0.826	0.558	0.560	0.589	0.647	0.775	0.785	0.758	0.865
Qwen3-235B	0.655	0.669	0.666	0.766	0.834	0.866	0.654	0.662	0.656	0.771	0.862	0.867	0.865	0.899
Llama-3.1-8B	0.532	0.536	0.534	0.632	0.642	0.685	0.442	0.448	0.485	0.599	0.655	0.668	0.612	0.664
Llama-3.1-70B	0.635	0.641	0.673	0.775	0.785	0.813	0.612	0.623	0.622	0.735	0.802	0.811	0.805	0.854
Llama-3.1-405B	0.642	0.655	0.718	0.812	0.822	0.854	0.703	0.715	0.711	0.820	0.858	0.865	0.861	0.891
GPT-5.5	0.642	0.655	0.736	0.841	0.855	0.891	0.724	0.738	0.733	0.846	0.884	0.891	0.887	0.923

Table 4: Prompt-matched and common-cap query-time controls on the full STAGE set ($N = 5010$) using Qwen3-235B. All variants use the same reasoner and skill-reader prompts, answer prompt, decoding configuration, maximum final-evidence budget, and bounded one-round supplement protocol. Calls (R/L/S) report the average number of Reasoner calls, Reader calls, and supplement actions per query, respectively; tool-internal model calls are excluded.

Variant	EvTok Mean	EvTok P95	Supp. Rate	Rounds	Calls (R/L/S)	Overall	Pass@5
HybridRAG + Agent + Skills	8,140	11,250	76.5%	0.77	1.0 / 1.0 / 1.45	0.628	0.692
LightRAG + Skills	7,620	10,840	72.8%	0.73	1.0 / 1.0 / 1.32	0.471	0.635
NKW w/o narrative assets	6,950	9,420	68.4%	0.68	1.0 / 1.0 / 1.24	0.648	0.742
Full NKW	7,210	9,650	64.2%	0.64	1.0 / 1.0 / 1.15	0.701	0.815

Table 5: Component ablations across benchmarks using Qwen3-235B. Gray subscripts indicate the absolute performance drop compared to the full system.

Variant	STAGE		FairytaleQA		QuALITY	
	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5
Full system	0.701	0.815	0.875	0.955	0.865	0.899
w/o attributes	0.671 $\downarrow 0.030$	0.772 $\downarrow 0.043$	0.851 $\downarrow 0.024$	0.921 $\downarrow 0.033$	0.817 $\downarrow 0.048$	0.871 $\downarrow 0.028$
w/o narrative hierarchy	0.648 $\downarrow 0.053$	0.742 $\downarrow 0.073$	0.860 $\downarrow 0.015$	0.931 $\downarrow 0.023$	0.789 $\downarrow 0.076$	0.842 $\downarrow 0.057$
w/o graph refinement	0.665 $\downarrow 0.036$	0.768 $\downarrow 0.047$	0.849 $\downarrow 0.026$	0.916 $\downarrow 0.039$	0.804 $\downarrow 0.061$	0.859 $\downarrow 0.040$
w/o snapshot-guided reasoner	0.641 $\downarrow 0.060$	0.735 $\downarrow 0.080$	0.845 $\downarrow 0.030$	0.905 $\downarrow 0.050$	0.775 $\downarrow 0.090$	0.829 $\downarrow 0.070$
w/o skill-based evidence assembly	0.625 $\downarrow 0.076$	0.705 $\downarrow 0.110$	0.835 $\downarrow 0.040$	0.885 $\downarrow 0.070$	0.755 $\downarrow 0.110$	0.809 $\downarrow 0.090$

points for 8B, 30B, and 235B, respectively. Thus, its contribution persisted even where the full system’s single-answer accuracy trailed Hybrid RAG.

5.3 Skill and Tool Analysis

Paired utility of bounded supplementation. To isolate the tool channel within skill-based evidence assembly, Q2 (skill-no-tool) and Q3 (skill-tool) reused the same question, snapshot preview, reasoner plan, and initial source packet.

Only Q3 allowed the selected skill to request a bounded supplement. A rescued question changed from incorrect in Q2 to correct in Q3; an induced error changed from correct to incorrect.

On STAGE, supplementation rescued 226 questions and introduced 45 errors, yielding a net gain of 3.6 percentage points (Table 6). Supplementation was requested for 64.2% of STAGE questions, compared with 14.5% on FairytaleQA and 8.3% on QuALITY; the latter two gained 0.7 and

Table 6: Paired utility and cost of bounded evidence supplementation with Qwen3-235B. Q2 and Q3 reuse the same question, snapshot preview, reasoner plan, and initial source packet; they differ only in whether the selected skill may request one bounded supplement. Δ Accuracy is the paired net change, i.e., rescued questions minus induced errors divided by the benchmark size.

Benchmark	Variant	Supplement Rate	Utility (Paired vs Q2)			Accuracy		Extra Latency	
			Rescued ($\uparrow$)	Errors ($\downarrow$)	Unchanged	Absolute	Δ Acc.	p50 (s)	p95 (s)
STAGE (N=5010)	Q2: skill-no-tool	0.0%	—	—	—	0.665	—	reference	reference
	Q3: skill-tool	64.2%	226	45	4739	0.701	+0.036	+8.5	+18.2
FairytaleQA (N=10580)	Q2: skill-no-tool	0.0%	—	—	—	0.868	—	reference	reference
	Q3: skill-tool	14.5%	120	46	10414	0.875	+0.007	+3.2	+7.5
QuALITY (N=2523)	Q2: skill-no-tool	0.0%	—	—	—	0.852	—	reference	reference
	Q3: skill-tool	8.3%	45	12	2466	0.865	+0.013	+2.8	+6.4

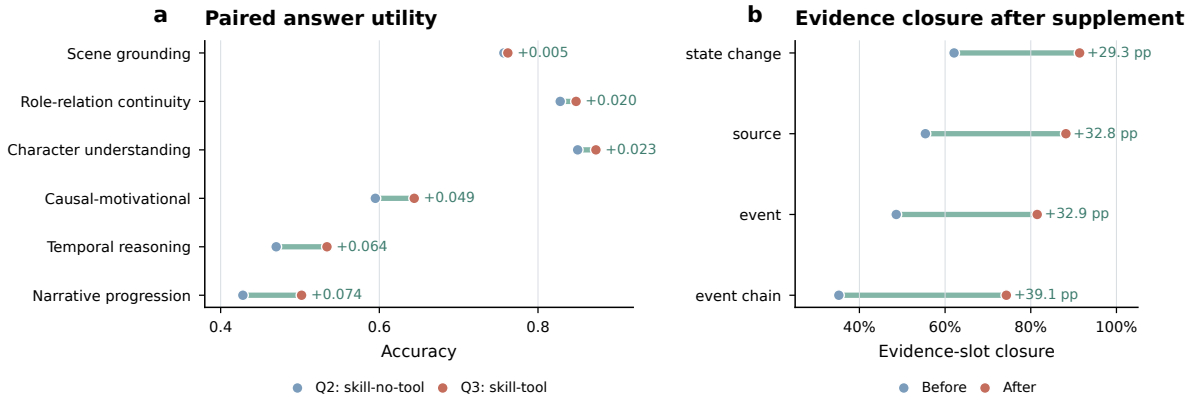


Figure 2: Paired utility and evidence closure for bounded supplementation with Qwen3-235B. (a) Dumbbells show Q2 (skill-no-tool) to Q3 (skill-tool) accuracy changes by STAGE question type. (b) The same supplement closes unresolved evidence slots across the four primary evidence capabilities; labels show the absolute closure gain in percentage points. Preview and plan composition are reported in Table C.1.

1.3 points, respectively. Figure 2 shows that the largest STAGE gains occurred in narrative progression, temporal, and causal reasoning. Evidence-slot closure also increased across the four primary evidence capabilities. Appendix C details preview selection, skill routing, source admission, and reliability.

Efficiency. The paired median additional latency was 8.5 seconds on STAGE, 3.2 on FairytaleQA, and 2.8 on QuALITY; corresponding p95 increases were 18.2, 7.5, and 6.4 seconds (Table 6). For overall system cost, the 40-screenplay diagnostic subset required 10.12 minutes per screenplay to build NKW’s assets and 36.45 seconds per query. LightRAG required 7.46 minutes and 9.15 seconds, while GraphRAG required 16.33 minutes and 485.50 seconds (Appendix B.9). DOS RAG averaged 8.88 seconds per query. These measurements place NKW’s accuracy gains alongside its additional construction and query overhead relative to lightweight retrieval. Downstream reuse of the narrative assets is illustrated in Appendix D.

6 Conclusion

We presented **Narrative Knowledge Weaver**, a framework for source-grounded narrative knowledge modeling and long-form QA. NKW aligns factual, graph, entity-profile, interaction, episode, and storyline layers so query-time reasoning can move between local evidence and higher-level narrative progression. It targets questions whose answers depend not on a single passage, but on how events, character states, relations, and plot threads evolve across the story. Results show that NKW is strongest on screenplay-level story-world QA, while remaining competitive on passage-centered benchmarks such as FairytaleQA and QuALITY, where lightweight retrieval can be preferable in some small-backbone or locally answerable settings. Overall, the findings suggest that narrative QA benefits from modeling how evidence functions inside a story, not only where relevant passages or entity relations occur, while future work should improve efficiency, stability, domain coverage, and source-grounded auditability.

Limitations

NKW is designed for long-form narrative understanding, and its advantages are strongest when questions require evidence across character states, events, temporal order, causal relations, and story progression. For shorter or more passage-centered questions, the additional narrative structure can be less beneficial than lightweight retrieval, as reflected by the strong performance of some baselines on FairytaleQA and QuALITY. The method also incurs higher construction and query-time cost than flat retrieval because it builds source-grounded narrative layers and performs multi-step evidence assembly.

The quality of the constructed asset bundle depends on upstream extraction and normalization. If events, entity identities, or source-grounded facts are missed or incorrectly merged, later retrieval and reasoning may inherit these errors. Although final answers are tied back to source evidence, the query-time agent can still select different evidence paths across runs, leading to variance in single-sample answers. We report the skill manifests and routing contract, but do not isolate the causal effect of each individual skill or study multi-skill interactions; the reported protocol intentionally selects one primary skill per query. Our evaluation focuses on QA over existing narrative datasets; we do not claim that the same configuration is sufficient for all narrative tasks, domains, or languages. The upgraded arc schema makes character goals, obstacles, turning points, and observed outcomes explicit, but it remains a compact three-level operationalization: it does not reconstruct every narrative level or infer an authorial global plot when the source leaves a goal or conflict unresolved. Such cases are retained as uncertain and should be expanded through source evidence rather than completed by the system.

Ethics Statement

This work uses existing scientific artifacts: public narrative and long-context QA datasets, language-model backbones, and retrieval or graph-RAG baselines. We cite the creators of the datasets and methods used in the experimental setup and related work. We follow the licenses or access terms of the original artifacts and use them for research evaluation; we do not redistribute original benchmark texts or proprietary model weights as part of this paper. The datasets consist of fictional or published narrative texts and benchmark questions; we do not

recruit human participants or annotators, collect new human-subject data, or collect personally identifying information from participants. Some source narratives may contain fictional violence, offensive language, stereotypes, or other sensitive content inherited from the original works. Our use is limited to research evaluation of narrative understanding systems.

The main risk of NKW is misuse as a tool for generating unsupported interpretations of narrative sources or for over-trusting automatically constructed story graphs. We mitigate this risk by keeping graph, factual, episode, and storyline records linked to source evidence and by requiring final answers to be grounded in original references. The system is intended for research and analysis settings rather than autonomous decision making about real people. We used AI assistants for research support, coding, and writing assistance; all paper content, experiments, and final claims were reviewed and edited by the authors.

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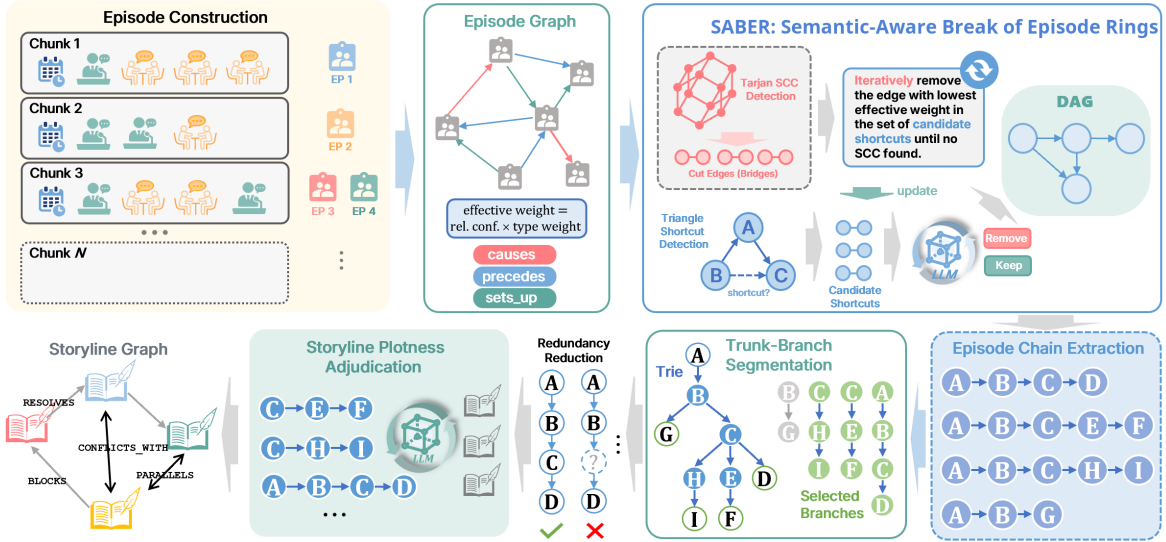


Figure A.1: **Three-level, theory-aligned narrative aggregation.** NKW assembles source-grounded events, interactions, and occasions into coherent episodes, cleans episode relations into an acyclic progression graph, and induces storyline arcs from bounded episode chains. Storyline records include goals, conflicts, outcomes, and links to supporting episodes. This is a compact operationalization of narrative structure (Piper et al., 2021).

- E.1 Artifacts, Licenses, and Intended Use (§E.1)
- E.2 Reproducibility and Decoding Settings (§E.2)

A Method Details and Query-Time Reasoning

This appendix collects method details that support the construction-time narrative hierarchy and the query-time QA pipeline. It first expands the episode/storyline aggregation procedure, then describes evidence assembly, query-time tool interfaces, evidence binding, and skill-based reading.

A.1 Narrative Aggregation Details

Figure A.1 expands the narrative aggregation module summarized in Section 3.2. The module starts from three kinds of base narrative units: *events*, which represent actions, discoveries, changes, conflicts, or plan outcomes; *interactions*, which represent directed exchanges among participants such as warning, deceiving, helping, threatening, persuading, or hiding information; and *occasions*, which represent stable narrative contexts such as scenes, social situations, institutional constraints, or recurring settings. These units provide the local evidence from which larger narrative structures are induced.

Episode assembly. An episode is a coherent mid-level narrative segment, not a single chunk label. It

may be local to one scene or span multiple sections when its units jointly express the same goal, conflict, action chain, information change, recurring occasion, or participant-centered situation. Candidate episode clusters are assembled using shared participants, shared goals or conflicts, causal dependency, temporal continuity, recurring occasions, semantic coherence, source-span adjacency, and explicit evidence links. Each episode records its child event, interaction, and occasion units, main participants, source spans, character-level goal states, goals, obstacles or conflicts, state changes, turning points, observed outcome, resolution status, and source-grounded evidence. This preserves the path from an episode back to the local units and passages that support it while separating an intended action from its observed result. Missing or ambiguous outcomes remain uncertain.

Episode relation graph and DAG cleaning. After episode assembly, NKW extracts relations among episodes. The main directed relations are *precedes*, *causes*, and *elaborates*; contrastive relations are kept as auxiliary semantic links rather than as main progression edges. Each relation has a confidence score and a type-dependent weight, producing an effective edge weight used during graph cleaning. The raw episode graph can contain cycles and redundant shortcuts. NKW therefore converts it into a clean episode DAG using SABER, a semantic-aware procedure for breaking cycles

and pruning redundant shortcuts in event-relation graphs. The procedure marks triangle shortcut candidates, repeatedly detects strongly connected components, removes the weakest edge inside each cyclic component while preferring edges already marked as redundant shortcuts, and then re-checks remaining shortcut candidates on the resulting DAG. Only the remaining ambiguous shortcuts are passed to an LLM adjudicator, which decides whether a direct edge is fully covered by an indirect path.

Algorithm 1 SABER Episode DAG Cleaning

Require: Noisy episode graph $G = (V, E)$; edge type $\tau(e)$; confidence $c(e)$; type weight w_τ
Ensure: Clean episode DAG G_{dag}

- 1: Compute effective weight $s(e) = c(e) \cdot w_{\tau(e)}$ for every edge.
- 2: Mark triangle-shortcut candidates (u, v) when a path $u \rightarrow x \rightarrow v$ exists.
- 3: **while** G contains a strongly connected component C with $|C| > 1$ **do**
- 4: Let $E_C = \{(u, v) \in E : u, v \in C\}$.
- 5: Remove the edge in E_C with minimum $s(e)$, preferring marked shortcut candidates.
- 6: Recompute strongly connected components.
- 7: **end while**
- 8: Recompute shortcut candidates on the acyclic graph.
- 9: **for all** ambiguous direct edge (u, v) with an indirect path $u \rightsquigarrow v$ **do**
- 10: Ask whether (u, v) is REDUNDANT, NECESSARY, or UNCERTAIN.
- 11: **if** the decision is REDUNDANT and source support is preserved **then**
- 12: Remove (u, v) .
- 13: **end if**
- 14: **end for**
- 15: **return** the resulting DAG G_{dag} .

Within SABER, the LLM is used only after the graph skeleton is acyclic. It cannot create episodes, add new edges, reverse episode order, or remove a support link unless the direct relation is judged redundant with respect to an already existing indirect path.

Chain extraction and storyline construction.

Storyline candidates are extracted from the clean episode DAG as episode chains. We use a trunk–branch view rather than relying only on a longest path: source-to-sink paths are enumerated under bounded depth and path-count constraints, inserted into a trie, and segmented into shared trunks and meaningful branches. This representation exposes common narrative prefixes and later divergences, which is useful for questions about plot alternatives, delayed consequences, and recurring arcs. A storyline is then induced from a selected episode

chain as a higher-level narrative arc with persistent character goals, obstacles or conflicts, goal progression, turning points, state changes, observed outcome, resolution status, impact, key characters, source spans, and child episode links. These fields capture how an arc develops without assuming that every goal succeeds or every conflict reaches closure. Storyline support edges link storylines to their episodes, whereas storyline relations describe high-level interactions among arcs, such as enabling, blocking, resolving, conflicting with, or paralleling another storyline.

Algorithm 2 Trunk–Branch Storyline Segmentation

Require: Clean episode DAG G_{dag} ; maximum chain length L ; overlap threshold ρ ; minimum new episodes m
Ensure: Storyline nodes and support edges

- 1: Enumerate bounded source-to-sink episode paths in G_{dag} .
- 2: Insert each path into a trie whose nodes are episode ids.
- 3: Extract candidate chains from shared trunks, branch continuations, and terminal paths.
- 4: Split candidates longer than L at branch or low-support cut points.
- 5: Rank candidates by coverage, coherence, source support, and new episode contribution.
- 6: Initialize selected chain set $S \leftarrow \emptyset$.
- 7: **for all** candidate chain P in ranked order **do**
- 8: **if** $\max_{Q \in S} \text{overlap}(P, Q) \leq \rho$ and P contributes at least m new episodes **then**
- 9: Add P to S .
- 10: **end if**
- 11: **end for**
- 12: **for all** $P \in S$ **do**
- 13: Induce a storyline with goals, conflicts, turning points, outcome, resolution status, source spans, and child episode ids.
- 14: Add support edges from the storyline to its child episodes.
- 15: **end for**
- 16: Predict closed-set storyline relations only between selected storylines.

The complete aggregation path is therefore: source evidence to base narrative units, base units to episodes, episodes to a clean episode DAG, and episode chains to storylines. This hierarchy is designed to support both high-level narrative retrieval and source-grounded answering: a query can retrieve a storyline or episode for narrative orientation, then follow support links back to the exact local units and passages needed for verification.

A.2 End-to-End QA Procedure and Tool Interface

Given a benchmark question and its associated document or screenplay, NKW answers the question through a preview-guided, source-grounded skill-

agent pipeline. The preview narrows navigation; the reasoner chooses evidence actions; the active skill governs interpretation and any bounded supplemental retrieval.

Pipeline overview.

1. **Snapshot preview.** Offline assets are projected into a small set of cards with stable identifiers, source spans, child links, participants, temporal/setting cues, character goals, obstacles, turning points, and outcome status. Alias, title, lexical, and vector indexes select cards for the question. Cards are planning hints and are never admitted as final evidence by themselves.
2. **One reasoner plan.** The same LLM reasoner receives the question, answer format, visible cards, and the source-grounding contract. It outputs validated constraints, evidence slots, explicit actions, and one primary skill. An action names the asset view, target card or source range, slot, and budget; missing or invalid actions are discarded rather than replaced by an implicit all-channel recall.
3. **Planned retrieval.** NKW executes only the validated actions. Each returned item carries a source identifier and is assigned to an evidence slot. Structured cards may guide expansion, but original passages and source-grounded records are the admissible support.
4. **Skill-based reading.** The reader loads the primary skill’s procedure and manifest. It fixes the question’s actor, scope, polarity, and temporal anchor, then compares the available source evidence using the skill’s narrative operation. The reader can submit an answer directly when the packet is sufficient.
5. **Conditional supplement.** If one unresolved evidence slot could change the answer, the active skill may request a bounded batch through the shared `retrieve_evidence` interface. The request must be neutral, source-grounded, and tied to the slot; at most one supplement round and two parallel evidence actions are allowed. No supplement is issued merely because a tool exists.
6. **Source-backed finalization.** The reader records findings with source references, marks unsupported or contradictory claims, and emits the final answer in the requested format.

Preview and plan schema. The current snapshot uses four card families. An entity card identifies a canonical referent and its source-linked profile; a claim card represents an atomic fact; a `narrative_unit` card represents an event, interaction, occasion, or scene; and a thread card represents an episode or storyline handle. All cards share an identifier, brief description, participants, temporal cue, provenance, child links, confidence, and uncertainty fields, with narrative arc fields populated when available.

A validated plan contains explicit evidence slots and actions. A slot records its identifier, the missing evidence need, competing answer tests when applicable, current support/contradiction lists, and a status. An action has an identifier, operation, asset view, target cards or source range, slots, query, and retrieval budget. The current operations are `read_sources`, `search`, and `follow_links`; these are first-pass routing actions, not yet tool calls. A skill-controlled `retrieve_evidence` request is counted separately as a bounded supplement.

Packet accounting. For each query we record the number of visible cards, selected cards, planned actions, requested and resolved source identifiers, and the initial packet counts for source passages, entities, relations, narrative items, and tool-returned text. We also record the packet token estimate before the reader call. This separates preview exposure and planned retrieval from the later supplement packet.

The protocol has two separate contracts. The preview/reasoner contract controls what is searched; the skill/reader contract controls how the returned evidence is interpreted. This prevents a first-pass reasoner from paying for every graph, fact, narrative, and vector channel while preserving a targeted route to missing evidence.

Entity-fact budget. Before an entity is shown in the evidence packet, NKW attaches query-relevant entity attributes and at most five atomic facts by default. Candidate atomic facts for the entity are ranked by question similarity, then only the top facts are retained. This keeps fine-grained source-grounded propositions available without allowing repeated local facts to dominate the prompt. We also implement dynamic and cluster-aware atomic-fact budgets, but the reported experiments use the stable per-entity top-5 setting.

Card family	Runtime role	Expansion / evidence link
entity	Canonical referent and entity profile	Source identifiers and linked attributes/facts
claim	Atomic source-grounded proposition	Source identifier for the supporting claim
narrative_unit	Event, interaction, occasion, or scene context	Source span and child narrative units
thread	Episode or storyline handle	Episode/storyline children and source support

Table A.1: Snapshot card families exposed to the reasoner. Cards guide navigation; final claims must be grounded in expanded source evidence.

Tool	Evidence view	Purpose
find_source_evidence	Source passages	Locate wording for a bounded evidence slot, preserving actor, scope, time, and polarity.
find_event_evidence	Events/interactions	Recover a source-grounded action, interaction, or state-changing event.
find_relation_evidence	Relations	Verify a directed or scoped relation between named entities.
find_state_change	Entity-state records	Retrieve before/after evidence for a character or entity at a requested time.
trace_event_chain	Event/episode links	Follow a bounded causal or temporal chain when a local passage does not distinguish cause and outcome.
locate_scene	Sections/scenes	Find the source region for a named setting, scene, or chapter context.
inspect_entity	Canonical entity profile	Resolve alias, identity, or source-linked attributes needed to bind a claim to the correct referent.

Table A.2: Evidence tools used by the reported NKW protocol. A skill can use only the subset assigned to its reading operation; the list is therefore a capability surface, not a mandatory retrieval sequence.

Tool interface. The query-time surface is a skill-permitted interface rather than a global menu exposed to every question. The active manifest grants only the operations that can resolve its evidence slots; all operations return source identifiers and are executed through one shared query budget.

The interface deliberately leaves channel selection to the reasoner and active skill. For example, `track-character-state` can inspect an entity and retrieve state transitions, whereas `resolve-narrative-perspective` can locate a scene and verify an attributed event. This keeps tools semantically meaningful and prevents duplicate aliases for the same low-level search.

A.3 Evidence Binding and Tool Policy

NKW separates navigation signals from admissible evidence. Preview cards, graph summaries, and narrative traces can point to a source, but the final reader grounds a finding in the original passage or a source-grounded record recovered through that pointer. When several assets point to the same source, their provenance is merged; when a compressed summary conflicts with the passage, the passage takes precedence. This policy matters for questions whose answer turns on exact wording, reported speech, scope, or option polarity.

The first pass is plan-driven. The reasoner may select source reads, event or relation lookups, state transitions, scene expansion, or link following, but an unselected channel is not implicitly run. Ev-

ery returned item is bound to an evidence slot and retained with its source identifier. A skill can request a supplement only for a named unresolved slot. The request must state the neutral fact to distinguish, use the original actors and event without assuming an option is true, and pass through the active manifest’s permission check. A shared budget permits at most one supplement round and two parallel actions; if the gap remains, the reader reports it rather than silently expanding retrieval.

This policy also distinguishes narrative traces from semantic source search. A trace follows source-grounded event, episode, and storyline links for temporal, causal, or trajectory questions. A source search locates the wording needed to verify a local claim, and a scene lookup expands the setting when the question depends on context. The reasoner and skill select among these operations from the question’s narrative constraints instead of executing all modes for every query.

A.4 Skill-Based Reading and Tool Control

NKW implements reading as a skill-based agent stage. A skill is a versioned procedure with a trigger, a narrative operation, source-grounding rules, and a manifest of permitted evidence tools. The reasoner binds one primary skill while forming the query plan; the skill then reads the returned source packet and may request only the bounded supplement allowed by its manifest. Thus, skills are part of the agent’s control loop rather than pas-

Narrative dimension	Runtime skill	Reading operation and permitted evidence
Agents, identity, and changing states	track-character-state	Bind the holder and temporal anchor, compare before/after state records, and preserve unknown information access. Uses source, state-change, event, and entity inspection tools.
Events, change, and causal outcome	trace-causes-and-outcomes	Distinguish action, intention, observed outcome, and causal link; follow a bounded event chain when a local passage is insufficient. Uses source, event, and event-chain tools.
Teller, focalizer, and attributed belief	resolve-narrative-perspective	Attribute an assertion to the narrator, speaker, or perceiver and keep it distinct from a world fact. Uses source, scene, and event evidence.
Source situation and local context	read-source	Read the original passage with actor, scope, time, negation, and reported-speech constraints fixed from the question. Uses source evidence and scene location.

Table A.3: Theory-aligned reading skills used by the reported NKW protocol. Each skill specifies its reading operation, completion contract, and allowed evidence tools so that the agent’s interpretation policy is auditable.

sive prompt decorations.

The skill catalog is grounded in the narrative dimensions used by our offline schema. [Piper et al. \(2021\)](#) discuss agents and change, event and causal structure, temporal and spatial situation, and point of view; Genette’s account of discourse order and perspective provides the corresponding narratological grounding ([Genette, 1983](#)). The table states the NKW operationalization. The mapping is a system design choice inspired by that theory, not a claim that the source paper specifies our tool calls or implementation details.

At inference time, the reasoner chooses a skill from the question, answer format, preview, and candidate evidence slots. The skill receives the original question and source passages together with the plan, so a preview goal cannot override the source or change the question’s temporal anchor. The reader first tries to resolve the slots already planned; it calls a permitted tool only when a particular missing fact could change the answer. Tool output is appended as source-linked findings and is checked by the same skill before finalization.

This design makes the theory-to-system boundary explicit. The offline extractor and hierarchy provide operational fields for agent, event, time, setting, perspective, and rationale. The runtime skills define how those fields are read for a question. Policies such as source precedence, neutral supplement queries and bounded budgets are NKW engineering contracts; they are not presented as direct consequences of narrative theory. Disabling the skill stage therefore removes the theory-guided reading operation and conditional tool control while leaving the source index and offline asset bundle intact.

B Evaluation Details and Additional Results

B.1 Benchmark Suite Overview

The main evaluation suite combines three public narrative and long-context QA resources. STAGE evaluates reasoning over full movie screenplays, providing one source of long, scene-structured narrative reasoning tasks within the broader benchmark suite. QuALITY evaluates multiple-choice reasoning over long passages where shallow retrieval is insufficient. FairytaleQA focuses on children’s stories and provides expert-authored questions over narrative elements such as characters, settings, actions, feelings, causal relations, and outcomes.

We report STAGE with semantic correctness metrics. QuALITY is reported with multiple-choice accuracy, and FairytaleQA is evaluated with exact or semantic answer correctness depending on the answer format used by the official split.

B.2 LLM-Evaluated Answer Correctness Prompt

For the primary semantic evaluation, we query DeepSeek-V4 once at temperature 0 for each (question, system answer, reference answer) triple. The same correctness prompt is used in a separate evaluator-agreement audit, where each item is judged five times with stochastic sampling. Those repeated judgments are used only to estimate pairwise agreement, Krippendorff’s α , and majority-vote consistency; they do not replace the deterministic scores reported in the main results.

Correctness Criterion. The evaluator judges correctness under the following definition:

- **Correctness** — An answer is considered correct if it preserves the key factual meaning of the reference answer, does not contradict core

facts, and does not introduce hallucinated or fabricated content. Minor variations in wording or level of detail are acceptable as long as the factual semantics match the reference answer.

The evaluator is instructed to be conservative: if an answer is only partially correct, omits critical information, or mixes correct facts with incorrect details, it should be labeled as incorrect.

LLM Correctness Evaluation Prompt Template

You are a careful factual evaluator. Your task is to determine whether the given answer is factually correct with respect to the reference answer. Judge correctness under the following rule:

- Label the answer as *correct* if it matches the key factual meaning of the reference answer without contradictions or hallucinations.
- Label the answer as *incorrect* if it contradicts the reference, misses essential information, adds fabricated details, or is only partially aligned.

Return a JSON object with the structure:

```
{
  "is_correct": true | false,
  "reason": "Short judgment explanation."
}
```

— Question —

{QUESTION}

— System Answer —

{SYSTEM_ANSWER}

— Reference Answer —

{REFERENCE_ANSWER}

Evaluate whether the system answer is factually correct with respect to the reference answer. Output only the JSON.

B.3 LLM Evaluator Agreement

To assess the stability of the LLM-based correctness evaluator (Appendix B.2), we conduct a separate agreement audit with five independent stochastic judgments for each (question, system answer, reference answer) triple. The repeated labels are used to compute pairwise agreement, Krippendorff’s α , and majority-vote consistency; the primary benchmark scores use the single deterministic judgment described above.

Pairwise agreement is computed over all $\binom{5}{2} = 10$ annotator pairs per example. To quantify multi-annotator consistency, we additionally report Krippendorff’s α for nominal-scale labels (Krippendorff, 2011). Let n_{ic} denote how many times label

$c \in \{0, 1\}$ appears among the five judgments for example i , and let n_c be the total number of occurrences of label c across the dataset. The observed and expected disagreements are

$$D_o = \frac{1}{N} \sum_{i=1}^N \frac{2n_{i0}n_{i1}}{5 \cdot 4},$$

$$D_e = \frac{2n_0n_1}{(n_0 + n_1)^2 - (n_0^2 + n_1^2)},$$

and Krippendorff’s α is

$$\alpha = 1 - \frac{D_o}{D_e}.$$

Because α measures how much the annotators agree beyond what would be expected by random labeling, it is well suited for evaluating the reliability of repeated stochastic LLM judgments.

Table B.1: Internal agreement of the correctness evaluator across five independent judgments. Pairwise agreement is averaged over all $\binom{5}{2} = 10$ label pairs per example. QuALITY is excluded because it is evaluated as multiple-choice accuracy rather than semantic answer correctness.

Dataset	Pairwise Agreement	Krippendorff’s α
STAGE	88.5%	0.76
FairytaleQA	91.2%	0.82

Most disagreements arise in borderline or partially correct answers, confirming that majority voting provides a robust and reliable estimate of semantic correctness.

B.4 Evaluator Stochasticity

To validate the reliability of our LLM-as-a-judge protocol (DeepSeek-V4), we run 10 independent evaluation passes on a random 200-question subset for each benchmark. Table B.2 reports the mean overall accuracy and standard deviation across systems with different generation paradigms. The results show tight stability: the largest standard deviation is 0.016, or 1.62 percentage points, across all settings.

The variance pattern is also interpretable rather than arbitrary. NKW has the lowest variance on both STAGE and FairytaleQA, consistent with its more structured and source-grounded answers leaving fewer borderline cases for the judge. GraphRAG and Hybrid RAG show slightly higher variance in some settings, which is expected when

abstractive summaries or flat-text retrieval produce partially correct, underspecified, or over-general answers. STAGE also shows marginally higher variance than FairytaleQA, reflecting the greater ambiguity of screenplay-level reasoning compared with more explicit fact-centric story questions. Overall, these small deviations support the stability of the evaluation protocol across distinct output formats and answer-quality regimes.

Table B.2: Stochasticity analysis of the LLM-as-a-judge evaluator.

Benchmark	System	Mean Acc.	Std (σ)
STAGE	NKW (Full)	0.702	0.011
	LightRAG	0.441	0.013
	GraphRAG	0.450	0.016
	Hybrid RAG	0.625	0.016
FairytaleQA	NKW (Full)	0.878	0.006
	LightRAG	0.875	0.007
	GraphRAG	0.449	0.010
	Hybrid RAG	0.841	0.009

B.5 STAGE Question-Type Breakdown

Table B.3 breaks down STAGE performance by its six annotated narrative reasoning types, with all methods using the Qwen3-235B backbone. The type distribution covers scene grounding, character understanding, causal-motivational reasoning, temporal reasoning, narrative progression, and role-relation continuity.

The breakdown shows that NKW improves most consistently on question types that require tracking narrative state across non-adjacent evidence. It is strongest on all six categories under the Qwen3-235B setting, with especially large margins on Temporal Reasoning and Causal-Motivational Reasoning. The absolute scores also reveal where the benchmark remains difficult: Temporal Reasoning and Narrative Progression remain relatively difficult, especially for systems without explicit narrative aggregation, indicating that long-range ordering and plot-trajectory questions remain challenging. Hybrid RAG remains competitive on local grounding and role-relation questions, but its Pass@5 gains are smaller than NKW’s on the more global categories, suggesting that narrative structure mainly helps when the model must assemble multiple constraints rather than retrieve a single salient passage.

B.6 STAGE Question-Type Ablation

The ablation results connect the component drops in Table 5 to the question types each component is designed to support. Removing attributes mainly hurts Character Understanding and Role-Relation Continuity, which depend on stable entity attributes, states, affiliations, and recurring relations. Removing narrative hierarchy has its largest effect on Temporal Reasoning, Causal-Motivational Reasoning, and Narrative Progression, showing that event-level evidence alone is insufficient for questions about ordering, motivation, and plot trajectory. Graph refinement also matters across categories because noisy episode relations can redirect long-range evidence. The two online ablations expose complementary effects: removing the snapshot-guided reasoner weakens initial evidence planning, while removing skill-based evidence assembly causes larger losses on causal, temporal, and progression-oriented questions where the retrieved evidence must be interpreted with the appropriate narrative operation. These results are separate from the paired bounded-supplement analysis in Appendix C.

B.7 Cross-Model Component Ablation

The following table reports the full STAGE component ablation across Qwen3-8B, Qwen3-30B, and Qwen3-235B. All variants use the same question sets, decoding configuration, and query-time budget as the corresponding full system.

B.8 STAGE Case Analysis

To complement the aggregate STAGE results, we inspect representative Qwen3-235B cases from *Birthday Girl*. This film is useful because many questions require separating adjacent domestic scenes, identifying the initiating event in a chain, or tracking a relationship shift across non-contiguous scenes. Table B.6 summarizes cases where NKW is correct in Pass@5 while several baselines produce majority-incorrect answers.

The cases expose three recurring advantages of the narrative-centric representation. First, scene-level provenance reduces local collisions: in Q2, the relevant decision is a short physical action inside the landing/bathroom sequence, and methods relying on less structured retrieval often substitute nearby but wrong moments. Second, event and causal traces help distinguish a cause from its consequence: in Q10, the answer is not merely that

Table B.3: STAGE question-type breakdown with all methods using Qwen3-235B. Bold marks the best method separately for Overall and Pass@5 within each question type.

Question Type	Count	Hybrid RAG		GraphRAG		LightRAG		HippoRAG		A-RAG		NKW (Ours)	
		Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5
Scene Grounding	1207	0.723	0.748	0.296	0.458	0.400	0.540	0.464	0.481	0.330	0.417	0.762	0.858
Character Understanding	1103	0.751	0.768	0.416	0.568	0.657	0.821	0.594	0.627	0.503	0.610	0.873	0.910
Causal-Motivational Reasoning	1041	0.534	0.570	0.199	0.283	0.403	0.562	0.413	0.454	0.342	0.474	0.644	0.765
Temporal Reasoning	766	0.362	0.454	0.260	0.435	0.279	0.435	0.163	0.180	0.266	0.450	0.534	0.736
Narrative Progression	597	0.446	0.499	0.360	0.623	0.333	0.489	0.163	0.176	0.189	0.310	0.502	0.688
Role-Relation Continuity	296	0.790	0.845	0.571	0.743	0.568	0.787	0.486	0.561	0.642	0.801	0.848	0.919
Overall	5010	0.606	0.646	0.321	0.479	0.441	0.599	0.401	0.430	0.362	0.486	0.701	0.815

Table B.4: Fine-grained STAGE question-type ablation with Qwen3-235B. Gray subscripts show absolute drops from the full system. The results separate local asset effects from structural aggregation and online planning effects.

(a) Overall Accuracy						
Variant	Character	Role-Relation	Scene	Causal-Motiv.	Temporal	Narrative Prog.
Full system	0.873	0.848	0.762	0.644	0.534	0.502
w/o attributes	0.807 _{↓0.066}	0.790 _{↓0.058}	0.742 _{↓0.020}	0.627 _{↓0.016}	0.521 _{↓0.013}	0.489 _{↓0.013}
w/o narrative hierarchy	0.857 _{↓0.016}	0.828 _{↓0.020}	0.733 _{↓0.029}	0.561 _{↓0.083}	0.407 _{↓0.127}	0.462 _{↓0.040}
w/o graph refinement	0.822 _{↓0.051}	0.784 _{↓0.064}	0.734 _{↓0.028}	0.610 _{↓0.034}	0.508 _{↓0.026}	0.474 _{↓0.029}
w/o snapshot-guided reasoner	0.833 _{↓0.040}	0.798 _{↓0.050}	0.704 _{↓0.058}	0.572 _{↓0.072}	0.449 _{↓0.085}	0.452 _{↓0.050}
w/o skill-based evidence assembly	0.823 _{↓0.050}	0.786 _{↓0.062}	0.692 _{↓0.070}	0.554 _{↓0.090}	0.414 _{↓0.120}	0.442 _{↓0.060}
(b) Pass@5						
Variant	Character	Role-Relation	Scene	Causal-Motiv.	Temporal	Narrative Prog.
Full system	0.910	0.919	0.858	0.765	0.736	0.688
w/o attributes	0.831 _{↓0.080}	0.838 _{↓0.081}	0.824 _{↓0.034}	0.735 _{↓0.030}	0.714 _{↓0.022}	0.667 _{↓0.022}
w/o narrative hierarchy	0.887 _{↓0.024}	0.882 _{↓0.037}	0.817 _{↓0.041}	0.661 _{↓0.104}	0.612 _{↓0.124}	0.558 _{↓0.131}
w/o graph refinement	0.852 _{↓0.058}	0.848 _{↓0.071}	0.814 _{↓0.044}	0.721 _{↓0.044}	0.700 _{↓0.037}	0.652 _{↓0.037}
w/o snapshot-guided reasoner	0.850 _{↓0.060}	0.849 _{↓0.070}	0.778 _{↓0.080}	0.675 _{↓0.090}	0.636 _{↓0.100}	0.608 _{↓0.080}
w/o skill-based evidence assembly	0.830 _{↓0.080}	0.829 _{↓0.090}	0.748 _{↓0.110}	0.635 _{↓0.130}	0.596 _{↓0.140}	0.578 _{↓0.110}

Table B.5: Core component ablations across model sizes on the full STAGE set. All variants use the same question sets, decoding configuration, and query-time budget as the corresponding full system. Gray subscripts indicate the absolute performance drop compared with the full system within the same model size.

Variant	Qwen3-8B		Qwen3-30B		Qwen3-235B	
	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5
Full system	0.385	0.501	0.589	0.664	0.701	0.815
w/o attributes	0.362 _{↓0.023}	0.478 _{↓0.023}	0.564 _{↓0.025}	0.638 _{↓0.026}	0.671 _{↓0.030}	0.772 _{↓0.043}
w/o narrative hierarchy	0.351 _{↓0.034}	0.463 _{↓0.038}	0.548 _{↓0.041}	0.612 _{↓0.052}	0.648 _{↓0.053}	0.742 _{↓0.073}
w/o graph refinement	0.370 _{↓0.015}	0.485 _{↓0.016}	0.569 _{↓0.020}	0.641 _{↓0.023}	0.665 _{↓0.036}	0.768 _{↓0.047}
w/o snapshot-guided reasoner	0.338 _{↓0.047}	0.445 _{↓0.056}	0.535 _{↓0.054}	0.598 _{↓0.066}	0.641 _{↓0.060}	0.735 _{↓0.080}
w/o skill-based evidence assembly	0.315 _{↓0.070}	0.422 _{↓0.079}	0.520 _{↓0.069}	0.585 _{↓0.079}	0.625 _{↓0.076}	0.705 _{↓0.110}

the ring is placed, but the preceding charged meal interaction that motivates it. Third, storyline-level structure helps temporal and progression questions choose the right abstraction level. In Q18 and Q28, several baselines retrieve events that are narratively related but occur at the wrong point in the chain; NKW more often anchors the answer to the initiating event required by the question.

We also inspect failure cases to clarify the current boundary of NKW’s reasoning behavior. Ta-

ble B.7 reports non-Scene-Grounding examples from diagnostic STAGE runs over *Birthday Girl* and *Chasing Amy*. These cases show that source-grounded narrative assets reduce many retrieval failures, but do not eliminate errors in selecting the exact causal trigger, temporal anchor, or speaker-level motivation when several semantically related events compete.

Table B.6: Representative STAGE cases from *Birthday Girl* under Qwen3-235B. The table summarizes majority-incorrect baseline behavior and contrasts it with NKW’s answers.

Case	Type	Reference	Baseline failure pattern	NKW behavior
Q2: bathroom encounter	Character Understanding	Nadia locks herself in with John.	LightRAG, HippoRAG, GraphRAG, Hybrid RAG, and A-RAG are majority-incorrect. They either report that no decision is evidenced, retrieve a different bathroom/landing moment, or hallucinate an emotional reaction such as Nadia sobbing.	NKW answers correctly in 4/5 runs by grounding the local action: Nadia closes the door and remains inside with John.
Q10: tense meal	Causal-Motivational Reasoning	Prolonged eye contact during the meal creates the tension that leads to the ring gesture.	HippoRAG and GraphRAG are majority-incorrect, while A-RAG over-focuses on the later ring placement itself. Incorrect answers often replace the silent tension with fabricated dialogue or with the consequence rather than the preceding cause.	NKW answers correctly in 5/5 runs by staying within the kitchen scene and connecting the ring gesture to the charged interaction immediately before it.
Q18: intimate trajectory	Narrative Progression	Nadia initiates physical intimacy after John’s rejection.	LightRAG, HippoRAG, GraphRAG, and Hybrid RAG are majority-incorrect. Typical errors shift the pivot to a violent fight, Alexei’s suite, or another broad plot event, losing the local agentive action that changes the trajectory.	NKW answers correctly in 4/5 runs, using the narrative causal trace to identify Nadia’s initiating action as the pivot toward the intimate encounter and its aftermath.
Q28: start of interaction	Temporal Reasoning	John greets arriving passenger Nadia at the airport before the motorway drive.	HippoRAG, Hybrid RAG, and A-RAG are majority-incorrect. Their answers drift to later travel events, such as a lift exchange, a car ignition, or leaving a police station, which are temporally adjacent but not the start of the interaction.	NKW is correct in Pass@5 by retrieving the airport interaction and distinguishing the initiating greeting from later movement toward the car.

Table B.7: Representative NKW failure cases from STAGE diagnostic runs. We exclude Scene Grounding questions to focus on reasoning failures rather than direct scene lookup.

Case	Type	Reference	NKW failure pattern	Diagnostic implication
<i>Birthday Girl</i> Q16	Narrative Progression	The airport meeting initiates the shift in John and Nadia’s relationship.	NKW repeatedly selects later intimate or domestic moments, such as the bedroom encounter or ring gesture, treating salient consequences as the initiating event.	Progression questions can still confuse the start of a trajectory with later high-salience events inside that trajectory.
<i>Chasing Amy</i> Q3	Character Understanding	Holden catches up to Alyssa in order to confront her about the rumors.	NKW answers from the broader romantic arc, saying that Holden seeks love or a relationship, rather than the local conversational objective in the parking-lot scene.	Character-intention questions remain vulnerable to global relationship summaries when the question asks for a local action goal.
<i>Chasing Amy</i> Q16	Narrative Progression	Banky’s mockery of Holden’s fixation on Alyssa escalates the tension between Holden and Banky.	NKW jumps to a later dramatic confrontation involving Holden’s threesome proposal and kiss, which is narratively related but not the earlier catalyst requested by the question.	Storyline retrieval can over-prefer dramatic downstream events unless the requested turning point is anchored to the correct causal position.
<i>Chasing Amy</i> Q33	Causal-Motivational Reasoning	Jay’s advice that Alyssa choosing Holden now matters more than her past motivates Holden to leave the diner.	NKW often attributes the decision to Silent Bob’s “chasing Amy” reflection or another nearby emotional cue, misidentifying both the speaker and the immediate trigger.	Dialogue-level causal attribution remains difficult when adjacent advice scenes share similar themes but differ in speaker, timing, and consequence.

B.9 STAGE Graph Asset Statistics

Table B.10 reports the graph-asset footprint on a stratified diagnostic subset of 40 screenplays drawn from the 151-screenplay STAGE pool. The 40 scripts were allocated across language and screenplay-length strata to approximately preserve the pool’s English/Chinese composition (109/42) and length distribution. This subset is used only for representation and runtime diagnostics; the main QA results use the full STAGE set. The numbers should be read as representation-scale statistics rather than as a direct quality metric, because each system defines nodes, edges, and auxiliary graph assets differently. Table B.11 further breaks down the NKW asset bundle into its entity-level and narrative-level components. Table B.12 reports the

corresponding construction and query-time wall-clock costs, measured under the same 16-worker parallel setup. Construction time is reported in minutes per screenplay, whereas QA + Retrieval is reported in seconds per query. SABER-specific ablations are reported separately in Appendix B.10.

Chunk-size sensitivity. The following diagnostic sweep rebuilds each index at four chunk sizes while keeping the model, answer prompt, decoding configuration, and query-time protocol fixed. Results are reported on the same stratified 40-screenplay subset used for representation diagnostics; they should not be read as replacement scores for the full 151-screenplay STAGE evaluation. QA sensitivity is summarized in Table B.8, and the corresponding graph-asset footprint is reported in Table B.9.

Table B.8: QA sensitivity to chunk size on a stratified 40-screenplay diagnostic subset using Qwen3-235B. Results are reported as Overall / Pass@5. Main comparisons in the paper use the 600-token operating point.

Chunk Size	Hybrid RAG	GraphRAG	LightRAG	HippoRAG	A-RAG	DOS RAG	NKW (Ours)
400	0.590 / 0.635	0.335 / 0.490	0.450 / 0.605	0.410 / 0.440	0.345 / 0.460	0.490 / 0.535	0.690 / 0.805
600	0.601 / 0.642	0.318 / 0.475	0.439 / 0.591	0.395 / 0.421	0.358 / 0.480	0.499 / 0.542	0.705 / 0.820
800	0.608 / 0.650	0.295 / 0.450	0.420 / 0.575	0.370 / 0.400	0.365 / 0.490	0.505 / 0.550	0.710 / 0.825
1200	0.595 / 0.640	0.270 / 0.410	0.405 / 0.550	0.350 / 0.380	0.370 / 0.500	0.510 / 0.555	0.695 / 0.810

Table B.9: Graph-asset stability across chunk sizes on the 40-screenplay subset. Text units, graph nodes and edges, and build time are reported as mean $\pm$ standard deviation per document. High-level asset entries report means; U, E, and S denote narrative units, episodes, and storylines. The table covers systems with explicit graph or high-level assets; Hybrid RAG and DOS RAG do not expose corresponding assets.

System	Chunk	Text Units	Graph Nodes	Graph Edges	High-level Assets	Build Time (min)
GraphRAG	400	245.2 $\pm$ 82.5	452.1 $\pm$ 115.4	1085.6 $\pm$ 240.2	95.4 $\pm$ 28.1 (Communities)	22.5 $\pm$ 8.2
	600	162.7 $\pm$ 55.4	397.8 $\pm$ 98.2	946.7 $\pm$ 215.5	87.9 $\pm$ 24.5 (Communities)	16.3 $\pm$ 5.8
	800	122.5 $\pm$ 41.2	345.5 $\pm$ 85.6	812.4 $\pm$ 180.4	75.2 $\pm$ 19.8 (Communities)	12.8 $\pm$ 4.5
	1200	81.8 $\pm$ 27.5	290.4 $\pm$ 72.1	655.2 $\pm$ 152.3	61.5 $\pm$ 16.2 (Communities)	9.4 $\pm$ 3.6
LightRAG	400	245.2 $\pm$ 82.5	645.2 $\pm$ 142.5	820.5 $\pm$ 188.4	–	10.2 $\pm$ 3.8
	600	162.7 $\pm$ 55.4	585.5 $\pm$ 125.8	752.2 $\pm$ 165.2	–	7.5 $\pm$ 2.9
	800	122.5 $\pm$ 41.2	512.4 $\pm$ 110.2	645.8 $\pm$ 140.5	–	6.1 $\pm$ 2.4
	1200	81.8 $\pm$ 27.5	435.8 $\pm$ 95.4	522.6 $\pm$ 115.8	–	4.5 $\pm$ 1.8
NKW (Ours)	400	245.2 $\pm$ 82.5	672.4 $\pm$ 155.2	890.5 $\pm$ 210.8	262 U, 84 E, 8.5 S	14.2 $\pm$ 5.1
	600	162.7 $\pm$ 55.4	624.5 $\pm$ 140.5	812.2 $\pm$ 195.4	245 U, 78 E, 8.4 S	10.1 $\pm$ 3.6
	800	122.5 $\pm$ 41.2	565.8 $\pm$ 125.6	735.6 $\pm$ 170.2	235 U, 76 E, 8.2 S	8.5 $\pm$ 3.0
	1200	81.8 $\pm$ 27.5	485.2 $\pm$ 105.4	625.4 $\pm$ 145.5	210 U, 68 E, 7.8 S	6.8 $\pm$ 2.4

Table B.10: STAGE graph-asset scale across graph-based systems. Statistics are per screenplay unless otherwise stated. Because graph schemas differ across systems, the table is intended as a footprint comparison rather than a direct measure of graph quality.

System	Asset	Mean	Median	Min	Max
HippoRAG	Nodes	1,252.0	1,236.0	244	2,886
	Edges	5,552.1	5,145.5	741	14,048
GraphRAG	Entity nodes	397.8	276.5	129	2,078
	Relationships	946.7	640.0	266	5,873
	Community reports	87.9	57.0	25	551
LightRAG	Nodes	585.5	574.0	229	1,020
	Edges	752.2	710.5	284	1,334
NKW (Ours)	KG nodes	624.5	595.0	245	1,150
	KG edges	812.2	780.5	310	1,480

Table B.11: Detailed NKW asset statistics on STAGE. NKW stores both a source-grounded entity graph and higher-level narrative assets, including episodes, storylines, and their support links.

Graph asset	Avg. / screenplay	Median	Min	Max
KG nodes	624.50	595.0	245	1,150
KG edges	812.20	780.5	310	1,480
Refined relations	835.40	795.5	315	1,520
Doc-entity edges	715.30	680.5	270	1,300
Documents	162.70	143.5	36	384
Episodes	78.50	75.0	30	140
Episode relations	118.20	112.5	45	210
Episode DAG relations	80.10	76.5	30	145
Storylines	8.35	8.0	3	15
Storyline relations	6.05	5.5	1	12
Storyline support edges	35.80	34.0	12	65

The comparison clarifies the representational difference behind the QA and ablation results. HippoRAG constructs the densest graph, especially in edge count. GraphRAG uses fewer entity nodes than NKW but maintains many relationships and community reports, while LightRAG keeps a smaller entity-relation footprint. NKW is not simply a larger flat graph: it maintains a sizeable entity graph together with document links, refined relations, episodes, storyline nodes, and cross-level support edges. This multi-level asset

bundle explains why the STAGE ablations in Table B.4 show different failure modes for local entity attributes, graph-channel tools, and episode/storyline aggregation. The intended benefit is not graph size alone, but separating local grounding from temporal, causal, and storyline-level evidence so the query-time agent can retrieve the right layer for each question type. The cost profile in Table B.12 shows the corresponding trade-off: NKW adds moderate construction overhead over LightRAG because it builds narrative hierarchy, and it

Table B.12: Computational cost for graph construction and query-time QA on STAGE.

System	Phase	Unit	Mean	Median	Min	Max
HippoRAG	Construction	min.	2.32	2.24	1.65	3.88
	QA + Retrieval	sec.	2.79	2.78	2.26	3.45
GraphRAG	Construction	min.	16.33	13.95	5.57	38.00
	QA + Retrieval	sec.	485.50	446.20	305.15	815.30
LightRAG	Construction	min.	7.46	6.66	2.96	16.84
	QA + Retrieval	sec.	9.15	8.80	5.92	18.75
NKW (Ours)	Construction	min.	10.12	8.95	3.85	22.50
	QA + Retrieval	sec.	36.45	34.20	18.50	82.50

incurs higher query-time latency because of agentic tool calls. It nevertheless remains far less expensive than GraphRAG’s community-summary query pipeline.

B.10 SABER Setting Ablation

We isolate the effect of SABER within the narrative aggregation pipeline. This ablation keeps episode assembly, query tools, reading skills, and the QA backbone (Qwen3-235B) fixed, varying only how raw episode relations are converted into the progression graph prior to chain extraction. **Raw** uses the uncleaned episode relation graph with bounded simple-path traversal to avoid infinite loops. **Heuristic DAG** removes the lowest-scoring edge inside each strongly connected component (SCC) until the graph is acyclic, but performs no semantic redundancy checks. **SABER** applies the full two-stage procedure: it first guarantees an acyclic skeleton, then uses LLM-late semantic pruning to remove direct edges that are already logically covered by indirect paths.

In the 151-screenplay STAGE construction pool, only 31 screenplays contained SCCs that strictly required cycle breaking. Tables B.13 and B.14 report this 31-screenplay SCC subset, where the difference between raw traversal, heuristic cycle breaking, and SABER is directly observable. Redundant semantic shortcuts, however, were pervasive across the broader corpus. SABER therefore applies LLM-adjudicated shortcut pruning regardless of whether the initial graph contains cycles, making it a general denoising step for storyline extraction. This comparison tests whether semantic-aware cycle breaking stabilizes the induced storyline asset bundle and whether cleaner trajectories translate into better QA on ordering-, causality-, and progression-sensitive questions.

The ablation separates two effects. The heuristic DAG removes the small number of cycle-inducing edges and improves over the raw graph, but it

still leaves many semantically redundant shortcuts. SABER removes those shortcuts after the graph skeleton is acyclic, yielding fewer progression edges, fewer storyline relations, and longer average storyline chains. The QA gains are concentrated on causal, temporal, and narrative-progression questions, which are precisely the categories that rely on clean episode order and non-spurious long-range trajectories.

C Query-Time Tool and Skill Diagnostics

This appendix reports the current query-time protocol. Each query first exposes a compact snapshot preview, then one reasoner plan, planned source retrieval, and one primary reading skill. A skill may request at most one bounded supplement through its permitted tools. All rates below use the stated benchmark or supplement-triggered-query denominator; Q2/Q3 utility is paired at the question level.

These diagnostics separate planned retrieval from conditional supplementation. The main text therefore reports only the paired utility and latency summary; the appendix exposes card selection, skill routing, source admission, slot closure, and reliability so that the evidence-assembly behavior can be audited.

D Downstream Applications

These case studies compose the offline narrative assets for application-specific analyses; they are not additional variants of the primary preview–reasoner–skill QA path.

D.1 Production Continuity Checking

This section describes a downstream application enabled by source-grounded scene–entity structure: *production continuity checking*. The goal is to determine whether two scenes can plausibly share a production setup. We do not assume a separate production-design extractor. Instead, the agent uses the same assets described in the main paper: source chunks, canonical entities, linked locations and objects, occasions, events, interactions, atomic facts, and scene titles. Visible production cues are considered only when they are supported by these source-grounded assets.

Overall pipeline. The continuity checker operates as an agent on top of the narrative KG and narrative stores. Rather than a fixed sequence of operations, the agent retrieves candidate scene pairs, builds a source-grounded evidence packet for each

Table B.13: SABER setting ablation on the 31 STAGE screenplays whose raw episode graphs contain SCCs. All settings reuse the same extracted episodes and raw episode relations; only the episode-relation cleanup strategy is changed before chain extraction and storyline construction. Asset statistics are averaged per screenplay, and QA metrics use the Qwen3-235B backbone.

Setting	Prog. Edges	Remain. SCC	Cycle Edges Removed	Shortcut Edges Removed	Storylines	Avg. SL Len.	SL Relations	Overall	Pass@5
Raw	363.7	0.32	0.0	0.0	12.4	6.2	14.8	0.671	0.772
Heuristic DAG	355.2	0.00	8.5	0.0	10.1	7.1	9.5	0.685	0.794
SABER	246.7	0.00	8.5	108.5	8.35	9.4	6.05	0.701	0.815

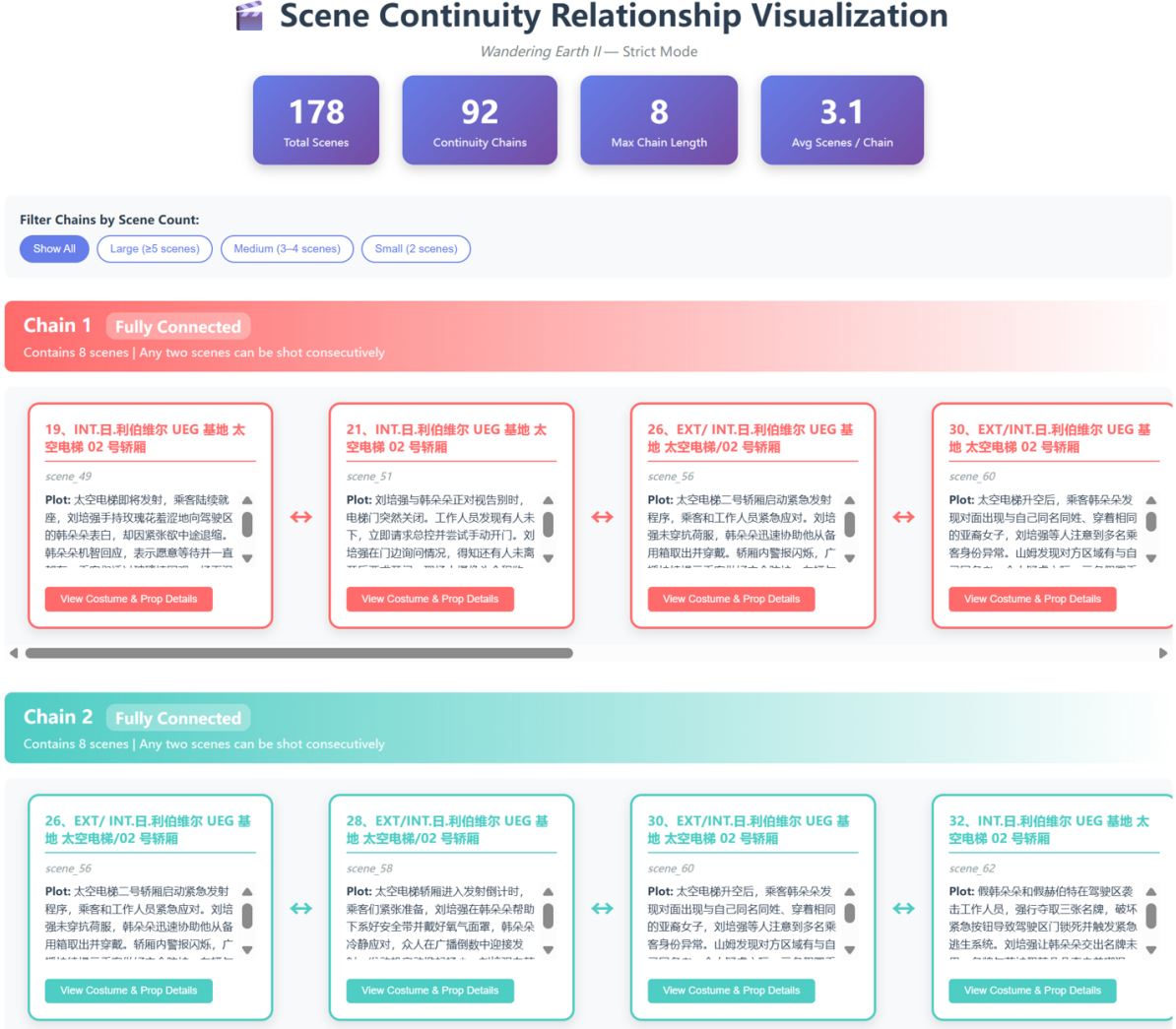


Figure D.1: **Continuity chain visualization.** Chains link scenes that can plausibly share a production setup in an example from *The Wandering Earth II* (Gwo, 2023), with scene summaries and source-grounded narrative cues shown for review.

scene, performs LLM-based judgments, and updates the continuity hypotheses. Each cycle consists of three core behaviors:

- **Candidate retrieval from the KG.** Querying all $O(N^2)$ scene pairs quickly becomes intractable as N grows. Instead, the agent first retrieves a sparse set of *candidate* scene pairs from the scene–entity graph. Two scenes are considered as candidates only if they share at

least one canonical entity node or occasion-linked entity, such as a character, location, or object. This simple structural prior already yields a substantial reduction in search space: for example, in *The Wandering Earth II* (Gwo, 2023) (173 scenes), naive all-pairs checking would require $173 \times 172/2$ pairwise decisions, whereas our entity-based filtering produces on the order of 1,500–2,000 candidates. The agent

Table B.14: SABER ablation on the SCC subset for STAGE question types most sensitive to episode ordering and storyline structure. Pruning semantic shortcuts improves temporal progression and causal-chain tracking.

Setting	Causal-Motiv.		Temporal		Narr. Prog.	
	Overall	Pass@5	Overall	Pass@5	Overall	Pass@5
Raw	0.611	0.712	0.472	0.665	0.445	0.612
Heuristic DAG	0.626	0.739	0.498	0.702	0.469	0.645
SABER	0.643	0.765	0.530	0.737	0.499	0.686

can further tighten this set by incorporating additional priors such as semantic similarity between scene summaries and constraints on script-order distance (e.g., only considering pairs whose scene indices differ by at most k).

- **LLM-based assessment and hypothesis update.** For each candidate pair, the agent invokes a structured continuity prompt conditioned on source-grounded scene packets: scene title/-source span, linked characters, locations, objects, occasions, shared entities, and relevant events, interactions, or atomic facts. The prompt asks whether the two scenes can share a production setup using only this evidence. The resulting binary decisions (with rationales) are used to update an evolving *continuity graph* over scenes, where edges represent hypothesized continuity relations.

- **Reflective chain-level refinement.** After accumulating pairwise judgments, the agent constructs preliminary continuity chains by extracting maximal cliques or connected components from the continuity graph. It then invokes a second LLM prompt—the *chain critic*—to reassess each chain holistically. The chain critic examines the entire sequence, identifies weakly supported edges, over-extended clusters, or source-grounded contradictions, and proposes merges, splits, or drops that refine the chain structure.

This iterative loop—KG-based candidate retrieval, LLM pairwise evaluation, and LLM chain-level reflection—constitutes the full continuity-checking agent. The continuity-chain visualization in Fig. D.1 is a presentation layer built on top of these outputs rather than a step in the agent itself.

Pairwise continuity prompt. We use the following prompt to assess whether two scenes belong to the same production setup.

Pairwise Production Continuity Judgment Prompt

Role: You are an expert reviewer evaluating whether two screenplay scenes can plausibly share the same *production setup*.

Definition (Production Continuity):

Two scenes are production-continuous if the provided source-grounded evidence supports reuse of a compatible physical setting, recurring objects or set dressing, and stable character/setup conditions. Narrative time jumps are allowed, but do not infer visual continuity unless the evidence supports it. Treat missing production details as unknown rather than hallucinating costumes, props, or lighting.

Scene 1

Title/source: {title1, chunks1}

Evidence: {source_evidence1}

Entities: {characters1, locations1, objects1, occasions1}

Facts: {events_interactions_facts1}

Scene 2

Title/source: {title2, chunks2}

Evidence: {source_evidence2}

Entities: {characters2, locations2, objects2, occasions2}

Facts: {events_interactions_facts2}

Shared evidence:

{shared_canonical_evidence}

Decision Rules

1. **Use only provided evidence.** Do not invent production-design details.
2. **Shared setup.** Favor continuity when scenes share a source-supported location, occasion, stable object, or recurring character configuration.
3. **Blocking evidence.** Reject continuity when the source indicates incompatible locations, mutually exclusive scene functions, or a changed state that requires a different setup.
4. **Uncertainty.** If evidence is sparse, lower confidence rather than fabricating a reason.

Return JSON fields:

```
{
  "is_continuity": true | false,
  "confidence": 0.0-1.0,
  "support": ["evidence"],
  "blockers": ["blocker"],
  "reason": "brief source-grounded reason"
}
```

LLM chain-reflection prompt. After extracting initial chains, an agent-style reflection step evaluates whether the chain is globally coherent and suggests refinements.

Table C.1: Preview and plan profile on STAGE. The initial packet is measured before any bounded supplement is requested. Card percentages are normalized within the visible or selected card set.

Statistic	Mean per Query	Card Distribution (Selected)				Initial Source IDs	Initial Packet Tokens
		Entity	Claim	Narrative	Thread		
Visible snapshot cards	46.8	31.5%	22.4%	36.2%	9.9%	12.4	6,540
Selected cards by reasoner	5.2	41.2%	12.5%	32.1%	14.2%		
Planned actions	4.1	—	—	—	—		
Generated evidence slots	3.4	—	—	—	—		

Table C.2: Skill routing and execution statistics on STAGE. Allowed tools is the number of tools permitted by the selected skill manifest; supplement rate is measured over queries assigned to that skill.

Skill Name	Selection Rate	Completion Rate	Protocol Rejections	Allowed Tools	Supplement Rate
read-source	34.2%	98.5%	12	2	18.4%
track-character-state	26.5%	96.2%	24	4	71.5%
trace-causes-and-outcomes	28.1%	95.8%	31	3	82.6%
resolve-narrative-perspective	11.2%	97.4%	8	3	55.2%

Table C.3: Evidence supplementation behavior on STAGE. Returned windows are candidates produced by the requested tool; admitted windows are source windows bound to an evidence slot and included in the final packet.

Primary Tool Requested	Calls / Query	Returned Windows	Admitted Windows	No-Match Rate	Slot Closure (Before)	Slot Closure (After)
find_source_evidence	1.15	4.2	1.8	8.5%	55.4%	88.2%
find_event_evidence	1.25	3.8	2.1	12.1%	48.6%	81.5%
trace_event_chain	1.40	5.1	2.5	14.5%	35.2%	74.3%
find_state_change	1.10	3.2	1.5	6.2%	62.1%	91.4%

Table C.4: Paired utility breakdown by STAGE question type with Qwen3-235B. Q2 is the skill reader without tools and Q3 enables bounded supplementation under the same reasoner plan.

Question Type	Questions	Utility (Paired vs Q2)			Accuracy		95% Paired CI
		Rescued (↑)	Errors (↓)	Unchanged	Q2	Q3	
Scene Grounding	1207	18	12	1177	0.757	0.762	[+0.001, +0.009]
Character Understanding	1103	35	10	1058	0.850	0.873	[+0.015, +0.031]
Causal-Motiv. Reasoning	1041	62	11	968	0.595	0.644	[+0.038, +0.060]
Temporal Reasoning	766	55	6	705	0.470	0.534	[+0.048, +0.080]
Narrative Progression	597	48	4	545	0.428	0.502	[+0.052, +0.096]
Role-Relation Continuity	296	8	2	286	0.828	0.848	[+0.005, +0.035]
Overall	5010	226	45	4739	0.665	0.701	[+0.029, +0.043]

Table C.5: Cost and reliability metrics per query on STAGE. Monetary cost uses the server-side price ledger for this run; token and time accounting are reported separately to make the overhead auditable.

Cost Metric	Value	Reliability Metric	Value
Reasoner model calls	1.00	Tool-internal fail count	0.02 / query
Reader model calls	1.00	Protocol rejection rate	1.5%
Tool-internal model calls	0.85	Source-binding error rate	2.1%
Total generation tokens (Out)	425	Slot closure failure rate	14.8%
Total context tokens (In)	11,250	Extra cost per rescued question	\$0.018

Continuity Chain Critic Prompt

Role: You are a senior reviewer assessing an automatically grouped *continuity chain*. Your task is to judge whether the chain is source-grounded and coherent as a reusable production setup.

Scenes: {ordered_scene_ids}

Scene packets:

{per_scene_source_grounding_packets}

Pairwise edges:

{pairwise_edge_decisions}

Evaluate:

- Whether every neighboring scene pair has explicit source-grounded support.
- Whether the chain depends on a weak bridge scene that should be split.
- Whether shared entities, locations, objects, and occasions remain compatible across the chain.

4. Whether any source-grounded fact contradicts reuse of the same setup.
5. Whether the chain is redundant, over-extended, or should be merged with another chain.

Return JSON fields:

```
{
  "coherence_score": 0.0-1.0,
  "confidence": 0.0-1.0,
  "decision": "keep" | "split" | "drop",
  "weak_edges": [
    ["scene_A", "scene_B"]
  ],
  "suggested_splits": [
    ["scene_A", "scene_B"],
    ["scene_C"]
  ],
  "rationale": "brief rationale"
}
```

D.2 Character State Tracking

This component derives fine-grained, per-scene character states from scene-structured narrative text. Whereas the main QA path assembles evidence for question answering, the character-state module targets a different layer of narrative grounding: short, observable descriptions of what important characters are doing in each scene.

Iterative extraction with reflection. For this downstream module, the agent performs a bounded *extract* $\rightarrow$ *reflect* $\rightarrow$ *revise* cycle. For each scene, the system (1) retrieves candidate characters from the scene-entity graph, (2) extracts only *visible*, *non-speculative* states using a structured LLM prompt, and (3) applies a critic model that checks coverage, grounding, and precision. Feedback is reinjected into the next extraction pass until convergence or a retry limit is reached. This loop yields a stable, scene-aligned collection of character states across the entire screenplay.

Observable state representation. Each character state captures only what is externally observable—actions, expressions, posture, and tone—excluding psychological or symbolic interpretation. Outputs use a minimal structured representation listing only important characters for each scene.

Visualization. To support interactive exploration, we render the extracted dataset using a timeline swimlane view.

1. Timeline Swimlane View. A global *character-scene matrix* showing all characters

(rows) across all scenes (columns). This view offers a high-level visualization of appearance frequency, co-occurrence structure, and participation patterns, enabling users to quickly identify which characters move together across the narrative. In a production setting, the same view can support assistant directors, continuity supervisors, and wardrobe/makeup teams by exposing character availability, scene grouping, co-presence, and state-continuity constraints across the shooting plan.

E Responsible NLP Checklist Details

E.1 Artifacts, Licenses, and Intended Use

Table E.1 summarizes the external benchmarks and compared systems used in the experiments. We use these artifacts for research evaluation only and do not redistribute original benchmark texts, screenplay sources, model weights, or API credentials with this paper. The Hybrid RAG baseline is implemented locally from standard lexical-semantic retrieval components and is therefore not listed as a separate external artifact.

E.2 Reproducibility and Decoding Settings

Table E.2 records the decoding and evaluator settings used for the experiments. Open-source backbone inference uses an 8 NVIDIA A100-SXM4-80GB GPU node (81,920 MiB per GPU) with NVIDIA driver 535.129.03. The exact number of GPUs used by a given model depends on the checkpoint size and tensor-parallel deployment, and GPU hours are not used as an evaluation metric.

For reproducibility, we plan to release the implementation, prompts, and configuration files needed to reproduce the reported experiments upon publication, subject to the licenses and access terms of the underlying datasets and model providers.

Query-time budget. The current NKW protocol exposes one shared query budget to the active skill. The reasoner-selected plan determines the initial actions; a reader may request at most one supplemental round with no more than two parallel evidence actions, and every action is bound to an evidence slot. Baselines use the shared 600-token chunking setup described in Section 4.

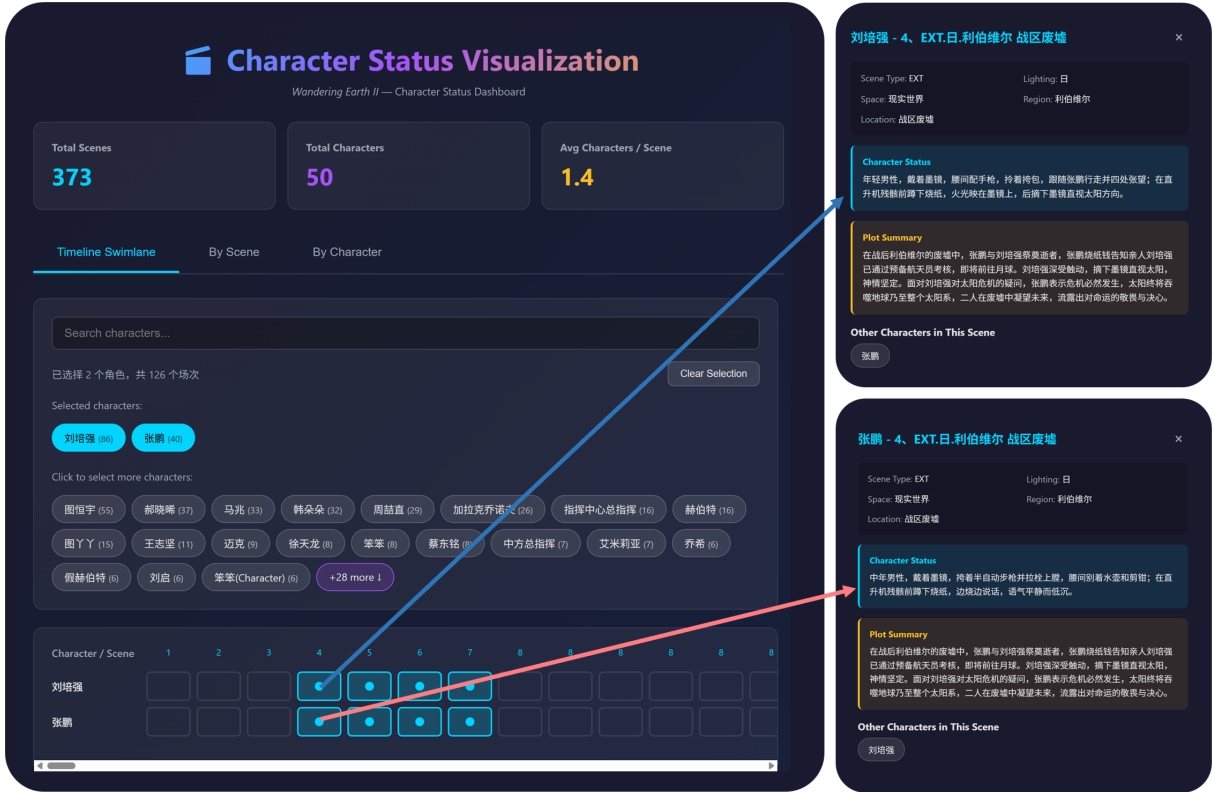


Figure D.2: **Timeline Swimlane View.** A screenplay-wide character–scene matrix enabling global inspection of appearance patterns.

Table E.1: External artifacts used in the experimental evaluation.

Artifact	Role	Source / citation	Use in this paper
STAGE	Benchmark	Tian et al. (2026)	Narrative and screenplay QA evaluation; raw screenplay text is not redistributed.
FairytaleQA	Benchmark	Xu et al. (2022)	Narrative QA evaluation over children’s stories.
QuALITY	Benchmark	Pang et al. (2022)	Long-input multiple-choice QA evaluation.
GraphRAG	Compared system	Edge et al. (2024)	Community-summary graph-RAG baseline.
LightRAG	Compared system	Guo et al. (2025)	Lightweight graph-augmented retrieval baseline.
HippoRAG	Compared system	Gutiérrez et al. (2024)	Associative graph-retrieval baseline.
A-RAG	Compared system	Du et al. (2026)	Agentic retrieval-interface baseline.

Table E.2: Reproducibility settings for generation and evaluation.

Stage	Model / interface	Temperature	Top- <i>p</i>	Max tokens	Notes
Open-source QA generation	Qwen3 / Llama-3.1 deployment	0.7	0.95	24,000	Samples five candidate answers per question for Pass@5; the higher temperature is used to avoid duplicate samples.
Closed-source QA generation	GPT-5.5 API	Provider default	Provider default	Provider default	Included as a closed-source reference under the provider’s standard access conditions.
Main semantic QA evaluation	DeepSeek-V4 API	0.0	1.0 or default	512	Deterministic LLM-as-a-judge pass; JSON output mode is used when supported by the provider.
Evaluator stochasticity test	DeepSeek-V4 API	0.1	1.0 or default	512	Ten independent evaluation passes on random 200-question subsets; a small nonzero temperature exposes borderline-answer variance.
Multiple-choice evaluation	Exact option matching	N/A	N/A	N/A	QuALITY is scored by selected-option accuracy rather than semantic judging.